Calculus I
Homework
Fermat's Theorem and the Mean Value Theorem

1. Evaluate the difference quotient and simplify your answer.

   (a) $\dfrac{f(2+h)-f(2)}{h}$, where $f(x) = x^2 - x - 1$.

   answer: $h + 3$

   (b) $\dfrac{f(a+h)-f(a)}{h}$, where $f(x) = x^2$.

   answer: $h + 2a$

   (c) $\dfrac{f(a+h)-f(a)}{h}$, where $f(x) = x^3$.

   answer: $h^2 + 3a^2 + 3ah$

   (d) $\dfrac{f(a+h)-f(a)}{h}$, where $f(x) = x^4$.

   answer: $6a^2h + 4ah^2 + h^3 + 4a^3$

   (e) $\dfrac{f(x)-f(a)}{x-a}$, where $f(x) = \frac{1}{x}$.

   answer: $-\frac{1}{ax}$.

   (f) $\dfrac{f(x)-f(1)}{x-1}$, where $f(x) = \frac{x-1}{x+1}$.

   answer: $\frac{1}{x+1}$.

2. Write down a formula for a function whose graphs is given below. The graphs are up to scale. Please note that there is more than one way to write down a correct answer.

(b) answer: $y = \begin{cases} -\frac{3}{2}x + 3 & \text{if } 0 < x \leq 2 \\ \frac{4}{3}x - \frac{8}{3} & \text{if } 2 < x \leq 6 \end{cases}$

(d) answer: $y = \begin{cases} -\frac{3}{4}x - \frac{1}{2} & \text{if } -2 \leq x < 2 \\ 5x - 12 & \text{if } 2 \leq x < 3 \end{cases}$

(c) answer: $y = \begin{cases} \frac{3}{4}x + \frac{5}{4} & \text{if } -3 \leq x < 1 \\ -4x + 6 & \text{if } 1 < x \leq 2 \end{cases}$

answer: $y = \begin{cases} -3x - 4 & \text{if } -2 \leq x < -1 \\ 2x + 1 & \text{if } -1 \leq x < 0 \\ -2x + 1 & \text{if } 0 \leq x < 1 \\ 3x - 4 & \text{if } 1 \leq x \leq 2 \end{cases}$

3. Write down formulas for function whose graphs are as follows. The graphs are up to scale. All arcs are parts of circles.

   (a)

4. Evaluate the difference quotient and simplify your answer.

   (a) $\dfrac{f(2+h) - f(2)}{h}$, where $f(x) = x^2 - x - 1$.

   answer: $h + 3$

   (b) $\dfrac{f(a+h) - f(a)}{h}$, where $f(x) = x^2$.

   answer: $h + 2a$

   (c) $\dfrac{f(a+h) - f(a)}{h}$, where $f(x) = x^3$.

   answer: $h^2 + 3a^2 + 3ah$

   (d) $\dfrac{f(a+h) - f(a)}{h}$, where $f(x) = x^4$.

   answer: $6a^2h + 4ah^2 + h^3 + 4a^3$

   (e) $\dfrac{f(x) - f(a)}{x - a}$, where $f(x) = \frac{1}{x}$.

   answer: $-\frac{1}{ax}$.

   (f) $\dfrac{f(x) - f(1)}{x - 1}$, where $f(x) = \frac{x-1}{x+1}$.

   answer: $\frac{1}{x+1}$.

5. Find the implied domain of the function.

   (a) $f(x) = \dfrac{x+4}{x^2 - 4}$.

   answer: $x \neq \pm 2$, alternatively: $x \in (-\infty, -2) \cup (-2, 2) \cup (2, \infty)$

   (b) $f(x) = \dfrac{2x^3 - 5}{x^2 + 5x + 6}$.

   answer: $x \neq -2, -3$, alternatively: $x \in (-\infty, -3) \cup (-3, -2) \cup (-2, \infty)$

   (c) $f(t) = \sqrt[3]{3t - 1}$.

   answer: $x \in \mathbb{R}$ (the domain is all real numbers)

   (d) $g(t) = \sqrt{5 - t} - \sqrt{1 + t}$.

   answer: $x \in [-1, 5]$.

   (e) $h(x) = \dfrac{1}{\sqrt[6]{x^2 - 7x}}$.

   answer: $x \in (-\infty, 0) \cup (7, \infty)$.

   (f) $f(u) = \dfrac{u+1}{1 + \frac{1}{u+1}}$.

   answer: $u \neq -1, -2$ or $u \in (-\infty, -2) \cup (-2, -1) \cup (-1, \infty)$..

   (g) $F(x) = \sqrt{10 - \sqrt{x}}$.

   answer: $x \in [0, 100]$

6. Find the implied domain of the function.

   (a) $f(x) = \dfrac{x+4}{x^2 - 4}$.

   answer: $x \neq \pm 2$, alternatively: $x \in (-\infty, -2) \cup (-2, 2) \cup (2, \infty)$

   (b) $f(x) = \dfrac{2x^3 - 5}{x^2 + 5x + 6}$.

   answer: $x \neq -2, -3$, alternatively: $x \in (-\infty, -3) \cup (-3, -2) \cup (-2, \infty)$

   (c) $f(t) = \sqrt[3]{3t - 1}$.

   answer: $x \in \mathbb{R}$ (the domain is all real numbers)

   (d) $g(t) = \sqrt{5 - t} - \sqrt{1 + t}$.

   answer: $x \in [-1, 5]$.

   (e) $h(x) = \dfrac{1}{\sqrt[6]{x^2 - 7x}}$.

   answer: $x \in (-\infty, 0) \cup (7, \infty)$.

   (f) $f(u) = \dfrac{u+1}{1 + \frac{1}{u+1}}$.

   answer: $u \neq -1, -2$ or $u \in (-\infty, -2) \cup (-2, -1) \cup (-1, \infty)$..

   (g) $F(x) = \sqrt{10 - \sqrt{x}}$.

   answer: $x \in [0, 100]$

7. Compute the composite functions $(f \circ g)(x)$, $(g \circ f)(x)$. Simplify your answer to a single fraction. Find the domain of the composite function.

   (a) $f(x) = \dfrac{x+2}{x-2}, g(x) = \dfrac{x-1}{x+2}$.

   answer: $(f \circ g)(x) = \frac{3+3x}{-5-x}$ $\quad x \neq -2, -5$; $(g \circ f)(x) = \frac{4}{-2+3x}$ $\quad x \neq 2, \frac{2}{3}$

   (b) $f(x) = \dfrac{x+1}{3x-2}, g(x) = \dfrac{x-2}{x-1}$.

   answer: $(f \circ g)(x) = \frac{-3+2x}{-4+x}$ $\quad x \neq 4, 1$; $(g \circ f)(x) = \frac{5-5x}{3-2x}$ $\quad x \neq \frac{2}{3}, \frac{3}{2}$

(c) $f(x) = \dfrac{2x+1}{3x-1}, g(x) = \dfrac{x-2}{2x-1}$.

answer: $(f \circ g)(x) = \frac{-5+4x}{-5+x}$ $x \neq 5, \frac{1}{2}$
$(g \circ f)(x) = \frac{3-4x}{3+x}$ $x \neq -3, \frac{1}{3}$

(d) $f(x) = \dfrac{x+1}{x-2}, g(x) = \dfrac{x+2}{2x-1}$.

answer: $(f \circ g)(x) = \frac{1+3x}{4-3x}$ $x \neq \frac{4}{3}, \frac{1}{2}$
$(g \circ f)(x) = \frac{-3+3x}{4+x}$ $x \neq -4, 2$

(e) $f(x) = \dfrac{5x+1}{4x-1}, g(x) = \dfrac{4x-1}{3x+1}$.

answer: $(f \circ g)(x) = \frac{-4+23x}{-5+13x}$ $x \neq \frac{5}{13}, -\frac{1}{3}$
$(g \circ f)(x) = \frac{5+16x}{2+19x}$ $x \neq -\frac{2}{19}, \frac{1}{4}$

(f) $f(x) = \dfrac{3x-5}{x-2}, g(x) = \dfrac{x-2}{x-4}$.

answer: $(f \circ g)(x) = \frac{-2x+14}{-x+6}$ $x \neq 6, 4$
$(g \circ f)(x) = \frac{x-1}{-x+3}$ $x \neq 3, 2$

(g) $f(x) = \dfrac{x-3}{x+2}, g(y) = \dfrac{y+3}{y-4}$.

answer: $(f \circ g)(x) = \frac{-2x+15}{3x-5}$ $x \neq \frac{5}{3}, 4$
$(g \circ f)(x) = \frac{4x+3}{-3x-11}$ $x \neq -\frac{11}{3}, -2$

8. Find the functions $f \circ g$, $g \circ f$, $f \circ f$ and $g \circ g$ and their implied domains. The answer key has not been proofread, use with caution.

(a) $f(x) = x^2 + 1, g(x) = x + 1$.

answer: Domain, all 4 cases: $x \in \mathbb{R}$ (all reals)
in some order: $(1+x)^2+1$, $(x)^2+2$, $((x)^2+1)^2+1$, $2+x$

(b) $f(x) = \sqrt{x+1}, g(x) = x + 1$.

answer: Domain of $f \circ g$ is $x \geq -2$. Domain of $g \circ f$ is $x \geq -1$
Domain of $f \circ f$ is $x \geq -1$. Domain of $g \circ g$ is all reals ($x \in \mathbb{R}$).
in some order: $\sqrt{2+x}$, $1+\sqrt{1+x}$, $\sqrt{1+\sqrt{1+x}}$, $2+x$

(c) $f(x) = 2x, g(x) = \tan x$.

In this subproblem, you are not required to find the domain.

answer: Domain $f \circ f$: all reals ($x \in \mathbb{R}$). Domain $g \circ f$: $x \neq (2k+1)\frac{\pi}{2}$ for all $k \in \mathbb{Z}$
Domain $f \circ g$: $x \neq (4k+1)\frac{\pi}{4}, x \neq (4k+3)\frac{\pi}{4}$ for all $k \in \mathbb{Z}$
Domain $g \circ g$: $x \neq (2k+1)\frac{\pi}{2}$ and $x \neq k\pi + \arctan\left(\frac{\pi}{2}\right)$ for all $k \in \mathbb{Z}$
in some order: $2\tan x$, $\tan(2x)$, $4x$, $\tan(\tan x)$

(d) $f(x) = \dfrac{x+1}{x-1}, g(x) = \dfrac{x-1}{x+1}$.

answer: Domain $f \circ f$: $x \neq 1$. Domain $g \circ g$: $x \neq 0, x \neq -1$
Domain $f \circ g$: $x \neq -1$. Domain $g \circ f$: $x \neq 0, x \neq 1$
in some order: $-x$, $\frac{1}{x}$, x, $-\frac{1}{x}$

9. Convert from degrees to radians.

(a) 15°.

answer: $\frac{\pi}{12} \approx 0.261799388$

(b) 30°.

answer: $\frac{\pi}{6} \approx 0.523598776$

(c) 36°.

answer: $\frac{\pi}{5} \approx 0.628318531$

(d) 45°.

answer: $\frac{\pi}{4} \approx 0.785398163$

(e) 60°.

answer: $\frac{\pi}{3} \approx 1.047197551$

(f) 75°.

answer: $\frac{5\pi}{12} \approx 1.308997$

(g) 90°.

answer: $\frac{\pi}{2}$

(h) 120°.

answer: $\frac{2\pi}{3}$

(i) 135°.

answer: $\frac{3\pi}{4}$

(j) 150°.

answer: $\frac{5\pi}{6}$

(k) 180°.

answer: π

(l) 225°.

answer: $\frac{5\pi}{4}$

(m) 270°.

answer: $\frac{3\pi}{2}$

(n) 305°.

answer: $\frac{61\pi}{36} \approx 5.323254$

(o) 360°.

answer: 2π

(p) 405°.

answer: $\frac{9\pi}{4}$

(q) 1200°.

answer: $\frac{20\pi}{3}$

(r) -900°.

answer: -5π

(s) -2014°.

answer: $-\frac{1007}{90}\pi \approx -35.150931$

10. Convert from radians to degrees. The answer key has not been proofread, use with caution.

(a) 4π.

answer: 720°

(b) $-\frac{7}{6}\pi$.

answer: -210°

(c) $\frac{7}{12}\pi$.

answer: 105°

(d) $\frac{4}{3}\pi$.

answer: 240°

(e) $-\frac{3}{8}\pi$.

answer: -67.5°

(f) 2014π.

answer: 362520°

(g) 5.

answer: $\left(\frac{900}{\pi}\right)^\circ \approx 286^\circ$

(h) -2014.

answer: -362520°

11. Prove the trigonometry identities.

(a) $\sin\theta\cot\theta = \cos\theta$.

(b) $(\sin\theta + \cos\theta)^2 = 1 + \sin(2\theta)$.

(c) $\sec\theta - \cos\theta = \tan\theta\sin\theta$.

(d) $\tan^2\theta - \sin^2\theta = \tan^2\theta\sin^2\theta$.

(e) $\cot^2\theta + \sec^2\theta = \tan^2\theta + \csc^2\theta$.

(f) $2\csc(2\theta) = \sec\theta\csc\theta$.

(g) $\tan(2\theta) = \dfrac{2\tan\theta}{1-\tan^2\theta}$.

(h) $\dfrac{1}{1-\sin\theta} + \dfrac{1}{1+\sin\theta} = 2\sec^2\theta$.

(i) $\tan\alpha + \tan\beta = \dfrac{\sin(\alpha+\beta)}{\cos\alpha\cos\beta}$.

(j) $\tan(\alpha+\beta) = \dfrac{\tan\alpha + \tan\beta}{1-\tan\alpha\tan\beta}$.

(k) $\sin(3\theta) + \sin\theta = 2\sin(2\theta)\cos\theta$.

(l) $\cos(3\theta) = 4\cos^3\theta - 3\cos\theta$.

(m) $1 + \tan^2\theta = \sec^2\theta$.

(n) $1 + \csc^2\theta = \cot^2\theta$.

(o) $2\cos^2(2x) = 2\sin^4\theta + 2\cos^4\theta - \sin^2(2\theta)$.

(p) $\dfrac{1+\tan\left(\frac{\theta}{2}\right)}{1-\tan\left(\frac{\theta}{2}\right)} = \tan\theta + \sec\theta$.

12. Find all values of x in the interval $[0, 2\pi]$ that satisfy the equation.

(a) $2\cos x - 1 = 0$.

answer: $x = \frac{\pi}{3}$ or $x = \frac{5\pi}{3}$

(b) $\sin(2x) = \cos x$.

answer: $x = \frac{\pi}{2}$, $x = \frac{3\pi}{2}$, $x = \frac{\pi}{6}$, or $x = \frac{5\pi}{6}$

(c) $\sqrt{3}\sin x = \sin(2x)$.

answer: $x = \frac{\pi}{6}, \frac{11\pi}{6}, 0, \pi, 2\pi$

(d) $2\sin^2 x = 1$.

answer: $x = \frac{\pi}{4}$, $x = \frac{3\pi}{4}$, $x = \frac{5\pi}{4}$, or $x = \frac{7\pi}{4}$

(e) $2 + \cos(2x) = 3\cos x$.

answer: $x = 0$, $x = 2\pi$, $x = \frac{\pi}{3}$, or $x = \frac{5\pi}{3}$

(f) $2\cos x + \sin(2x) = 0$.

answer: $x = \frac{\pi}{2}$, $x = \frac{3\pi}{2}$

(g) $2\cos^2 x - (1+\sqrt{2})\cos x + \dfrac{\sqrt{2}}{2} = 0$.

answer: $x = \frac{\pi}{4}, \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{4}$

(h) $|\tan x| = 1$.

answer: $x = \frac{\pi}{4}$, $x = \frac{3\pi}{4}$, $x = \frac{5\pi}{4}$, or $x = \frac{7\pi}{4}$

(i) $3\cot^2 x = 1$.

answer: $x = \frac{\pi}{3}$, $x = \frac{2\pi}{3}$, $x = \frac{4\pi}{3}$, or $x = \frac{5\pi}{3}$

(j) $\sin x = \tan x$.

answer: $x = 0$, $x = \pi$, or $x = 2\pi$

Solution. 12.g Set $\cos x = u$. Then

$$2\cos^2 x - (1+\sqrt{2})\cos x + \frac{\sqrt{2}}{2} = 0$$

becomes

$$2u^2 - (1+\sqrt{2})u + \frac{\sqrt{2}}{2} = 0.$$

This is a quadratic equation in u and therefore has solutions

$$\begin{aligned} u_1, u_2 &= \frac{1+\sqrt{2} \pm \sqrt{(1+\sqrt{2})^2 - 4\sqrt{2}}}{4} \\ &= \frac{1+\sqrt{2} \pm \sqrt{1 - 2\sqrt{2} + 2}}{4} \\ &= \frac{1+\sqrt{2} \pm \sqrt{(1-\sqrt{2})^2}}{4} \\ &= \frac{1+\sqrt{2} \pm (1-\sqrt{2})}{4} = \left\{ \begin{array}{l} \frac{1}{2} \quad \text{or} \\ \frac{\sqrt{2}}{2} \end{array} \right. \end{aligned}$$

Therefore $u = \cos x = \frac{1}{2}$ or $u = \cos x = \frac{\sqrt{2}}{2}$, and, as x is in the interval $[0, 2\pi]$, we get $x = \frac{\pi}{3}, \frac{5\pi}{3}$ (for $\cos x = \frac{1}{2}$) or $x = \frac{\pi}{4}, \frac{7\pi}{4}$ (for $\cos x = \frac{\sqrt{2}}{2}$).

13. Evaluate the limits. Justify your computations.

(a) $\lim\limits_{x\to 2} 2x^2 - 3x - 6.$ answer: -4

(b) $\lim\limits_{x\to -1} \dfrac{x^4 - x}{x^2 + 2x + 3}.$ answer: 1

(c) $\lim\limits_{x\to -1} \dfrac{1}{x^2 - 3x + 2}.$ answer: $\frac{1}{6}$

(d) $\lim\limits_{x\to -2} \sqrt{x^4 + 16}.$ answer: $\sqrt{32}$

(e) $\lim\limits_{x\to 8} (1 + \sqrt[3]{x})(2 - x).$ answer: -18

14. Evaluate the limit if it exists.

(a) $\lim\limits_{x\to 2} \dfrac{x^2 - 5x + 6}{x - 2}.$ answer: -1

(b) $\lim\limits_{x\to 3} \dfrac{x^2 - 3x}{x^2 - 2x - 3}.$ answer: $\frac{3}{4}$

(c) $\lim\limits_{x\to -2} \dfrac{2x^2 + x - 6}{x^2 - 4}$ answer: $\frac{7}{4}$

(d) $\lim\limits_{x\to 2} \dfrac{x^2 - 5x - 6}{x - 2}.$ answer: DNE

(e) $\lim\limits_{x\to -1} \dfrac{x^2 - 3x}{x^2 - 2x - 3}.$ answer: DNE

(f) $\lim\limits_{x\to -2} \dfrac{x^2 - 4}{2x^2 + 5x + 2}.$ answer: $\frac{4}{3}$

(g) $\lim\limits_{x\to -1} \dfrac{2x^2 + 3x + 1}{3x^2 - 2x - 5}.$ answer: $\frac{1}{8}$

(h) $\lim\limits_{x\to -4} \dfrac{x^2 + 7x + 12}{x^2 + 6x + 8}.$ answer: $\frac{1}{2}$

(i) $\lim\limits_{h\to 0} \dfrac{(-3 + h)^2 - 9}{h}.$ answer: -6

(j) $\lim\limits_{h\to 0} \dfrac{(-2 + h)^3 + 8}{h}.$ answer: 12

(k) $\lim\limits_{x\to -3} \dfrac{x + 3}{x^3 + 27}.$ answer: $\frac{1}{27}$

(l) $\lim\limits_{x\to 1} \dfrac{x^4 - 1}{x^3 - 1}.$ answer: $\frac{4}{3}$

(m) $\lim\limits_{h\to 0} \dfrac{\sqrt{4 + h} - 2}{h}.$ answer: $\frac{1}{4}$

(n) $\lim\limits_{x\to 3} \dfrac{\sqrt{5x + 1} - 4}{x - 3}.$ answer: $\frac{5}{8}$

(o) $\lim\limits_{x\to -3} \dfrac{\sqrt{x^2 + 16} - 5}{x + 3}.$ answer: $-\frac{3}{5}$

(p) $\lim\limits_{x\to -3} \dfrac{\frac{1}{3} + \frac{1}{x}}{3 + x}.$ answer: $-\frac{1}{9}$

(q) $\lim\limits_{x\to -2} \dfrac{x^2 + 4x + 4}{x^4 - 16}.$ answer: 0

(r) $\lim\limits_{x\to 0} \dfrac{\sqrt{1 + x} - \sqrt{1 - x}}{x}.$ answer: 1

(s) $\lim\limits_{x\to 0} \left(\dfrac{1}{x} - \dfrac{1}{x^2 + x}\right).$ answer: 1

(t) $\lim\limits_{x\to 9} \dfrac{3 - \sqrt{x}}{9x - x^2}.$ answer: $\frac{1}{54}$

(u) $\lim\limits_{h\to 0} \dfrac{(2 + h)^{-1} - 2^{-1}}{h}.$ answer: $-\frac{1}{4}$

(v) $\lim\limits_{x\to 0} \left(\dfrac{1}{x\sqrt{1 + x}} - \dfrac{1}{x}\right).$ answer: $-\frac{1}{2}$

(w) $\lim\limits_{h\to 0} \dfrac{(x + h)^3 - x^3}{h}.$ answer: $3x^2$

(x) $\lim\limits_{h\to 0} \dfrac{\frac{1}{(x+h)^2} - \frac{1}{x^2}}{h}.$ answer: $-\frac{2}{x^3}$

(y) $\lim\limits_{h\to 0} \dfrac{\frac{1}{(2+h)^2} - \frac{1}{4}}{h}.$ answer: $-\frac{1}{4}$

(z) $\lim\limits_{h\to 0} \dfrac{\frac{1}{(1+h)^2} - 1}{h}.$ answer: -2

Solution. 14.a

$$\begin{array}{rcl|l} \lim\limits_{x\to 2} \dfrac{x^2 - 5x + 6}{x - 2} &=& \lim\limits_{x\to 2} \dfrac{(x - 3)\cancel{(x - 2)}}{\cancel{x - 2}} & \text{factor and cancel} \\ &=& 2 - 3 = -1 & \end{array}$$

Solution. 14.c

$$\begin{array}{rcl|l} \lim\limits_{x\to -2} \dfrac{2x^2 + x - 6}{x^2 - 4} &=& \lim\limits_{x\to -2} \dfrac{(2x - 3)\cancel{(x + 2)}}{(x - 2)\cancel{(x + 2)}} & \text{factor and cancel} \\ &=& \dfrac{(2(-2) - 3)}{-2 - 2} & \text{substitute} \\ &=& \dfrac{7}{4} & \end{array}$$

Solution. 14.f

$$\begin{aligned}\lim_{x\to 2}\frac{x^2-4}{2x^2+5x+2} &= \lim_{x\to -2}\frac{(x-2)\cancel{(x+2)}}{(2x+1)\cancel{(x+2)}} && \Big|\ \text{factor and cancel}\\ &= \frac{(-2)-2}{2(-2)+1}=\frac{4}{3}.\end{aligned}$$

Solution. 14.g

$$\begin{aligned}\lim_{x\to -1}\frac{2x^2+3x+1}{3x^2-2x-5} &= \lim_{x\to -1}\frac{(2x+1)\cancel{(x+1)}}{(3x-5)\cancel{(x+1)}} && \Big|\ \text{factor and cancel}\\ &= \frac{2(-1)+1}{3(-1)-5}=\frac{1}{8}.\end{aligned}$$

Solution. 14.h.

$$\begin{aligned}\lim_{x\to -4}\frac{x^2+7x+12}{x^2+6x+8} &= \lim_{x\to -4}\frac{(x+3)\cancel{(x+4)}}{(x+2)\cancel{(x+4)}} && \Big|\ \text{factor}\\ &= \frac{-4+3}{-4+2}=-\frac{1}{2}.\end{aligned}$$

Solution. 14.x

$$\begin{aligned}\lim_{h\to 0}\frac{\frac{1}{(x+h)^2}-\frac{1}{x^2}}{h} &= \lim_{h\to 0}\frac{x^2-(x+h)^2}{hx^2(x+h)^2}=\lim_{h\to 0}\frac{x^2-(x^2+2xh+h^2)}{hx^2(x+h)^2}\\ &= \lim_{h\to 0}\frac{\cancel{h}(-2x+h)}{\cancel{h}x^2(x+h)^2}=\frac{-2x+0}{x^2(x+0)^2}=-\frac{2}{x^3}.\end{aligned}$$

Solution. 14.y.

Variant I.

$$\begin{aligned}\lim_{h\to 0}\frac{\frac{1}{(2+h)^2}-\frac{1}{4}}{h} &= \lim_{h\to 0}\frac{\frac{4-(2+h)^2}{4(2+h)^2}}{h}\\ &= \lim_{h\to 0}\frac{4-(4+4h+h^2)}{4h(2+h)^2}\\ &= \lim_{h\to 0}\frac{-4h-h^2}{4h(2+h)^2}\\ &= \lim_{h\to 0}\frac{\cancel{h}(-4-h)}{4\cancel{h}(2+h)^2} && \Big|\ \text{substitute } h=0\\ &= \frac{-4-0}{4(2+0)^2}\\ &= -\frac{1}{4}\end{aligned}$$

Variant II.

$$\begin{aligned}\lim_{h\to 0}\frac{\frac{1}{(2+h)^2}-\frac{1}{4}}{h} &= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{x^2}\right)_{|x=2}\\ &= \left(\frac{-2}{x^3}\right)_{|x=2}\\ &= -\frac{1}{4}\end{aligned}$$

Solution. 14.z.

Variant I.

$$
\begin{aligned}
\lim_{h\to 0}\frac{\frac{1}{(1+h)^2}-1}{h} &= \lim_{h\to 0}\frac{\frac{1-(1+h)^2}{(1+h)^2}}{h}\\
&= \lim_{h\to 0}\frac{1-(1+2h+h^2)}{h(1+h)^2}\\
&= \lim_{h\to 0}\frac{-2h-h^2}{h(1+h)^2}\\
&= \lim_{h\to 0}\frac{\cancel{h}(-2-h)}{\cancel{h}(1+h)^2} && \Big|\ \text{substitute } h=0\\
&= \frac{-2-0}{(1+0)^2}\\
&= -2.
\end{aligned}
$$

Variant II.

$$
\begin{aligned}
\lim_{h\to 0}\frac{\frac{1}{(1+h)^2}-1}{h} &= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{x^2}\right)_{|x=1} && \Big|\ \text{derivative definition}\\
&= \left(\frac{-2}{x^3}\right)_{|x=1}\\
&= -2.
\end{aligned}
$$

15. Find the (implied) domain of $f(x)$. Extend the definition of f at $x=3$ to make f continuous at 3.

(a) $f(x)=\dfrac{x^2-x-6}{x-3}$.

answer: Implied domain: $x\in(-\infty,3)\cup(3,\infty)$. Extend $f(x)$ to $f(x)=x+2$.

(b) $f(x)=\dfrac{x^3-27}{x^2-9}$.

answer: Implied domain: $x\in(-\infty,-3)\cup(-3,3)\cup(3,\infty)$. Extend $f(x)$ to $f(x)=\frac{x^2+3x+9}{x+3}$ with domain $x\in(-\infty,-3)\cup(-3,\infty)$.

16. Use the Intermediate Value Theorem to show that there is a real number solution of the given equation in the specified interval.

(a) $x^5+x-3=0$ where $x\in(1,2)$.

(b) $\sqrt[4]{x}=1-x$ where $x\in\mathbb{R}$ (i.e., x is an arbitrary real number).

(c) $\cos x=2x$, where $x\in(0,1)$.

(d) $\sin x=x^2-x-1$, where $x\in\mathbb{R}$ (i.e., x is an arbitrary real number).

(e) $\cos x=x^4$, where $x\in\mathbb{R}$ (i.e., x is an arbitrary real number).

(f) $x^5-x^2+x+3=0$, where $x\in\mathbb{R}$.

17.

(a) i. Solve the equation $x^2+13x+41=1$.

ii. Use the intermediate value theorem to prove that the equation $x^2+13x+41=\sin x$ has at least two solutions, lying between the two solutions to 17.a.i.

(b) i. Solve the equation $x^2-15x+55=1$.

ii. Use the intermediate value theorem to prove that the equation $x^2-15x+55=\cos x$ has at least two solutions, lying between the two solutions to the equation in the preceding item.

Solution. 17.a.i.

$$
\begin{aligned}
x^2+13x+41 &= 1\\
x^2+13x+40 &= 0\\
(x+5)(x+8) &= 0 \quad .
\end{aligned}
$$

eqnarray Therefore the two solutions are $x_1=-5$ and $x_2=-8$.

17.a.ii. Consider the function

$$f(x)=x^2+13x+41-\sin x \quad .$$

Our strategy for proving $f(x)=0$ has a solution consists in finding a number a such that $f(a)<0$ and a number b such that $f(b)>0$, and then using the Intermediate Value Theorem (IVT) with $N=0$.

Let

$$g(x)=x^2+13x+41,$$

and so $f(x) = g(x) - \sin x$. We have no techniques for evaluating $\sin x$ without calculator, but we do have all knowledge necessary to evaluate $g(x)$. Indeed, from high school we know that the lowest point of the parabola $g(x)$ is located at $x = -\frac{13}{2} = -6.5$. Then $g(-6.5) = -1.25$. Therefore

$$f(-6.5) = g(-6.5) - \sin(-6.5) = g(-6.5) + \sin(6.5) = -1.25 + \sin 6.5 \leq -0.25,$$

where for the very last inequality we use the fact that $\sin 6.5 < 1$ (remember $\sin t \leq 1$ for all real values of t).

On the other hand,

$$f(-5) = g(-5) - \sin(-5) = 1 + \sin 5 > 0$$

as $\sin 5 > -1$ (remember $\sin t \geq -1$ for all real values of t). Therefore $f(-5) > 0$ and $f(-6.5) < 0$ and by the Intermediate Value Theorem (IVT) $f(x) = 0$ has a solution in the interval $x \in (-6.5, -5)$.

Proving $f(x) = 0$ has a solution in the interval $x \in (-8, -6.5)$ is similar and we leave it to the student.

Below is a computer generated plot of the function with the use of which we can visually verify our answer.

18. For which values of x is f continuous?

- $f(x) = \begin{cases} 0 & \text{if } x \text{ is rational} \\ 1 & \text{if } x \text{ is irrational} \end{cases}$
- $f(x) = \begin{cases} 0 & \text{if } x \text{ is rational} \\ x & \text{if } x \text{ is irrational} \end{cases}$

19. Show that $f(x)$ is continuous at all irrational points and discontinuous at all rational ones.

$$f(x) = \begin{cases} \frac{1}{q^2} & \text{if } x \text{ is rational and } x = \frac{p}{q} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$$

where in the first item p, q are relatively prime integers (i.e., integers without a common divisor).

20. Show the following limits do not exist and compute whether they evaluate to ∞, $-\infty$, or neither.

(a) $\lim\limits_{x\to 3^+} \dfrac{x^2+x-1}{x^2-2x-3}$. answer: ∞.

(b) $\lim\limits_{x\to 3^-} \dfrac{x^2+x-1}{x^2-2x-3}$. answer: $-\infty$.

(c) $\lim\limits_{x\to 1^+} \dfrac{x^2+1}{\sqrt{x^2+3}-2}$. answer: ∞.

(d) $\lim\limits_{x\to 1^-} \dfrac{x^2+1}{\sqrt{x^2+3}-2}$. answer: $-\infty$.

(e) $\lim\limits_{x\to 2^+} \dfrac{\sqrt{x^3-8}}{-x^2+x+2}$. answer: $-\infty$.

(f) $\lim\limits_{x\to -1^+} \dfrac{\sqrt[3]{x^2+2x+1}}{x^2-2x-3}$. answer: $-\infty$.

21. Find the limit or show that it does not exist. If the limit does not exist, indicate whether it is $\pm\infty$, or neither. The answer key has not been proofread, use with caution.

(a) $\lim\limits_{x\to\infty} \dfrac{x-2}{2x+1}$. answer: $\frac{1}{2}$

(b) $\lim\limits_{x\to\infty} \dfrac{1-x^2}{x^3-x-1}$. answer: 0

(c) $\lim\limits_{x\to-\infty} \dfrac{x-2}{x^2+5}$. answer: 0

(d) $\lim\limits_{x\to-\infty} \dfrac{3x^3+2}{2x^3-4x+5}$. answer: $\frac{3}{2}$

(e) $\lim\limits_{x\to\infty} \dfrac{\sqrt{x}+x^2}{\sqrt{x}-x^2}$. answer: -1

(f) $\lim\limits_{x\to\infty} \dfrac{3-x\sqrt{x}}{2x^{\frac{3}{2}}-2}$. answer: $-\frac{1}{2}$

(g) $\lim\limits_{x\to\infty} \dfrac{(2x^2+3)^2}{(x-1)^2(x^2+1)}$. answer: 4

(h) $\lim\limits_{x\to\infty} \dfrac{x^2-3}{\sqrt{x^4+3}}$. answer: 1

(i) $\lim\limits_{x\to-\infty} \dfrac{\sqrt{x^2+1}}{x+1}$. answer: -1

(j) $\lim\limits_{x\to\infty} \dfrac{\sqrt{16x^6-3x}}{x^3+2}$. answer: 4

(k) $\lim\limits_{x\to-\infty} \dfrac{\sqrt{16x^6-3x}}{x^3+2}$. answer: -4

(l) $\lim\limits_{x\to\infty} \dfrac{\sqrt{3x^2+2x+1}}{x+1}$. answer: $\sqrt{3}$

(m) $\lim\limits_{x\to\infty} \sqrt{4x^2+x}-2x$. answer: $\frac{1}{4}$

(n) $\lim\limits_{x\to-\infty} x+\sqrt{x^2+3x}$. answer: $-\frac{3}{2}$

(o) $\lim\limits_{x\to\infty} \sqrt{x^2+2x}-\sqrt{x^2-2x}$. answer: 2.

(p) $\lim\limits_{x\to-\infty} \sqrt{x^2+x}-\sqrt{x^2-x}$. answer: -1.

(q) $\lim\limits_{x\to\infty} \sqrt{x^2+ax}-\sqrt{x^2+bx}$. answer: $\frac{a-b}{2}$

(r) $\lim\limits_{x\to\infty} \cos x$. answer: DNE

(s) $\lim\limits_{x\to\infty} \dfrac{x^4+x}{x^3-x+2}$. answer: ∞

(t) $\lim\limits_{x\to\infty} \sqrt{x^2+1}$. answer: ∞

(u) $\lim\limits_{x\to-\infty} (x^4+x^5)$. answer: $-\infty$

(v) $\displaystyle\lim_{x\to-\infty} \frac{\sqrt{1+x^6}}{1+x^2}$.

answer: ∞

answer: DNE

(x) $\displaystyle\lim_{x\to\infty} (x^2 - x^3)$.

answer: ∞

answer: $-\infty$

(z) $\displaystyle\lim_{x\to\infty} \sqrt{x}\sin x$.

(w) $\displaystyle\lim_{x\to\infty} (x - \sqrt{x})$.

(y) $\displaystyle\lim_{x\to\infty} x\sin x$.

answer: DNE

Solution. 21.d.

$$\begin{aligned}
\lim_{x\to-\infty} \frac{3x^3+2}{2x^3-4x+5} &= \lim_{x\to-\infty} \frac{\left(3x^3+2\right)\frac{1}{x^3}}{\left(2x^3-4x+5\right)\frac{1}{x^3}} \\
&= \lim_{x\to-\infty} \frac{3+\frac{2}{x^3}}{2-\frac{4}{x^2}+\frac{5}{x^3}} \\
&= \frac{3+0}{2-0+0} = \frac{3}{2}.
\end{aligned}
\qquad \left|\ \begin{array}{l}\text{Divide top}\\ \text{and bottom}\\ \text{by highest term}\\ \text{in denominator}\end{array}\right.$$

Solution. 21.i

$$\begin{aligned}
\lim_{x\to-\infty} \frac{\sqrt{x^2+1}}{x+1} &= \lim_{x\to-\infty} \frac{\frac{1}{x}\sqrt{x^2+1}}{\frac{1}{x}(x+1)} = \lim_{x\to-\infty} \frac{-\frac{1}{\sqrt{x^2}}\sqrt{x^2+1}}{\frac{1}{x}(x+1)} \qquad \left|\ x = -\sqrt{x^2},\ \text{whenever } x<0\right. \\
&= \lim_{x\to-\infty} \frac{-\sqrt{\frac{x^2+1}{x^2}}}{1+\frac{1}{x}} = \lim_{x\to-\infty} \frac{-\sqrt{1+\underbrace{\frac{1}{x^2}}_{\to 0}}}{1+\underbrace{\frac{1}{x}}_{x\to 0}} \\
&= 1.
\end{aligned}$$

Solution. 21.k.

$$\begin{aligned}
\lim_{x\to-\infty} \frac{\sqrt{16x^6-3x}}{x^3+2} &= \lim_{x\to-\infty} \frac{\sqrt{x^6\left(16-\frac{3}{x^5}\right)}}{x^3+2} \\
&= \lim_{x\to-\infty} \frac{\sqrt{x^6}\sqrt{\left(16-\frac{3}{x^5}\right)}}{x^3+2} \qquad \left|\ \sqrt{x^6} = -x^3 \text{ because } x<0 \text{ as } x\to-\infty\right. \\
&= \lim_{x\to-\infty} \frac{-x^3\sqrt{\left(16-\frac{3}{x^5}\right)}}{x^3+2} \qquad \left|\ \text{Divide by highest order term in denominator}\right. \\
&= \lim_{x\to-\infty} \frac{-x^3\sqrt{\left(16-\frac{3}{x^5}\right)}}{x^3+2} \\
&= \lim_{x\to-\infty} \frac{-\cancel{x^3}\sqrt{\left(16-\frac{3}{x^5}\right)}\frac{1}{\cancel{x^3}}}{\left(x^3+2\right)\frac{1}{x^3}} \\
&= \lim_{x\to-\infty} \frac{-\sqrt{\left(16-\underbrace{\frac{3}{x^5}}_{\to 0}\right)}}{1+\underbrace{\frac{2}{x^3}}_{\to 0}} \\
&= \frac{-\sqrt{16}}{1} = -4.
\end{aligned}$$

Solution. 21.l

$$
\begin{aligned}
\lim_{x\to\infty}\frac{\sqrt{3x^2+2x+1}}{x+1} &= \lim_{x\to\infty}\frac{\frac{1}{x}\sqrt{3x^2+2x+1}}{\frac{1}{x}(x+1)}\\
&= \lim_{x\to\infty}\frac{\sqrt{\frac{3x^2+2x+1}{x^2}}}{\left(1+\frac{1}{x}\right)}\\
&= \lim_{x\to\infty}\frac{\sqrt{3+\frac{2}{x}+\frac{1}{x^2}}}{\left(1+\frac{1}{x}\right)}\\
&= \frac{\sqrt{3+0+0}}{1+0}\\
&= \sqrt{3}.
\end{aligned}
$$

Solution. 21.p.

$$
\begin{aligned}
\lim_{x\to-\infty}\sqrt{x^2+x}-\sqrt{x^2-x} &= \lim_{x\to-\infty}\left(\sqrt{x^2+x}-\sqrt{x^2-x}\right)\frac{\left(\sqrt{x^2+x}+\sqrt{x^2-x}\right)}{\left(\sqrt{x^2+x}+\sqrt{x^2-x}\right)}\\
&= \lim_{x\to-\infty}\frac{x^2+x-(x^2-x)}{\sqrt{x^2+x}+\sqrt{x^2-x}} = \lim_{x\to-\infty}\frac{2x\frac{1}{x}}{\left(\sqrt{x^2+x}+\sqrt{x^2-x}\right)\frac{1}{x}}\\
&= \lim_{x\to-\infty}\frac{2}{\frac{\sqrt{x^2+x}}{x}+\frac{\sqrt{x^2-x}}{x}} = \lim_{x\to-\infty}\frac{2}{-\sqrt{\frac{x^2+x}{x^2}}-\sqrt{\frac{x^2-x}{x^2}}}\\
&= \lim_{x\to-\infty}\frac{2}{-\sqrt{1+\frac{1}{x}}-\sqrt{1-\frac{1}{x}}} = \frac{2}{-\sqrt{1+0}-\sqrt{1-0}} = -1.
\end{aligned}
$$

The sign highlighted in red arises from the fact that, for negative x, we have that $x=-\sqrt{x^2}$.

22. Find the horizontal and vertical asymptotes of the graph of the function. If a graphing device is available, check your work by plotting the function.

(a) $y=\dfrac{2x}{\sqrt{x^2+x+3}-3}$.

answer: Vertical: $x=2$, $x=-3$, horizontal: $y=2$, $y=-2$

(b) $y=\dfrac{3x^2}{\sqrt{x^2+2x+10}-5}$.

answer: Vertical: $x=3$, $x=-5$, horizontal: none.

(c) $y=\dfrac{3x+1}{x-2}$.

answer: vertical: $x=2$, horizontal: $y=3$

(d) $y=\dfrac{x^2-1}{2x^2-3x-2}$.

answer: vertical: $x=2$, $x=-\frac{1}{2}$, horizontal: $y=\frac{1}{2}$

(e) $y=\dfrac{2x^2-2x-1}{x^2+x-2}$.

answer: vertical: $x=1$, $x=-2$, horizontal: $y=2$

(f) $f(x)=\dfrac{-5x^2-3x+5}{x^2-2x-3}$

answer: vertical: $x=-1$, $x=3$, horizontal: $y=-5$

(g) $y=\dfrac{1+x^4}{x^2-x^4}$.

answer: vertical: $x=0$, $x=1$, $x=-1$, horizontal: $y=-1$

(h) $y=\dfrac{x^3-x}{x^2-7x+6}$.

answer: vertical: $x=6$, no horizontal asymptote

(i) $y=\dfrac{x-9}{\sqrt{4x^2+3x+3}}$.

answer: no vertical asymptote, horizontal: $y=\pm\frac{1}{2}$

(j) $y=\dfrac{\sqrt{x^2+1}-x}{x}$.

answer: vertical: $x=0$, horizontal: $y=0$, $y=-2$

(k) $f(x)=\dfrac{x}{\sqrt{x^2+3}-2x}$

answer: vertical: $x=1$, horizontal: $y=-\frac{1}{3}$, $y=-1$

Solution. 22.a **Vertical asymptotes.** A function $f(x)$ has a vertical asymptote at $x=a$ if $\lim\limits_{x\to a}f(x)=\pm\infty$.

The function is algebraic, and therefore has a finite limit at every point it is defined (i.e., no asymptote). Therefore the function can have vertical asymptotes only for those x for which $f(x)$ is not defined. The function is not defined for $\sqrt{x^2+x+3}-3=0$, which has two solutions, $x=2$ and $x=-3$. These are precisely the vertical asymptotes: indeed,

$$
\lim_{x\to2^+}\frac{2x}{\sqrt{x^2+x+3}-3}=\infty \qquad \lim_{x\to2^-}\frac{2x}{\sqrt{x^2+x+3}-3}=-\infty
$$

and

$$
\lim_{x\to-3^+}\frac{2x}{\sqrt{x^2+x+3}-3}=\infty \qquad \lim_{x\to-3^-}\frac{2x}{\sqrt{x^2+x+3}-3}=-\infty
$$

Horizontal asymptotes. A function $f(x)$ has a horizontal asymptote if $\lim\limits_{x\to\pm\infty}f(x)$ exists. If that limit exists, and is some number, say, N, then $y=N$ is the equation of the corresponding asymptote.

Consider the limit $x \to -\infty$. We have that

$$\begin{aligned}
\lim_{x\to-\infty} \frac{2x}{\sqrt{x^2+3x+3}-3} &= \lim_{x\to-\infty} \frac{2}{\frac{\sqrt{x^2+x+3}}{x} - \frac{3}{x}} \\
&= \lim_{x\to-\infty} \frac{2}{-\sqrt{\frac{x^2+3x+3}{x^2}} - \frac{3}{x}} && \left|\; \frac{1}{x} = -\sqrt{\frac{1}{x^2}} \text{ when } x<0 \right. \\
&= \lim_{x\to-\infty} \frac{2}{-\sqrt{1+\frac{3}{x}+\frac{3}{x^2}} - \frac{3}{x}} \\
&= \frac{\lim\limits_{x\to-\infty} 2}{-\sqrt{\lim\limits_{x\to-\infty} 1 + \lim\limits_{x\to-\infty} \frac{3}{x} + \lim\limits_{x\to-\infty} \frac{3}{x^2}} - \lim\limits_{x\to-\infty} \frac{3}{x}} \\
&= \frac{2}{-\sqrt{1+0+0}-0} \\
&= -2 \quad .
\end{aligned}$$

Therefore $y=-2$ is a horizontal asymptote.

The case $x \to \infty$, is handled similarly and yields that $y = 2$ is a horizontal asymptote.

A computer generated graph confirms our computations.

Solution. 22.d

Vertical asymptotes. A function $f(x)$ has a vertical asymptote at $x=a$ if $\lim\limits_{x\to a} f(x) = \pm\infty$.

The function is algebraic, and therefore has a finite limit at every point it is defined (i.e., no asymptote). Therefore the function can have vertical asymptotes only for those x for which $f(x)$ is not defined. The function is not defined for $2x^2-3x-2=0$, which has two solutions, $x=2$ and $x=-\frac{1}{2}$. These are precisely the vertical asymptotes: indeed,

$$\begin{aligned}
\lim_{x\to 2^+} \frac{x^2-1}{2x^2-3x-2} &= \lim_{x\to 2^+} \frac{x^2-1}{2(x-2)\left(x+\frac{1}{2}\right)} = \infty && \left|\; \text{Limit of form } \frac{(+)}{(+)(+)} \right. \\
\lim_{x\to 2^-} \frac{x^2-1}{2x^2-3x-2} &= \lim_{x\to 2^-} \frac{x^2-1}{2(x-2)\left(x+\frac{1}{2}\right)} = -\infty && \left|\; \text{Limit of form } \frac{(+)}{(-)(+)} \right.
\end{aligned}$$

and

$$\begin{aligned}
\lim_{x\to -\frac{1}{2}^+} \frac{x^2-1}{2x^2-3x-2} &= \lim_{x\to -\frac{1}{2}^+} \frac{x^2-1}{2(x-2)\left(x+\frac{1}{2}\right)} = \infty && \left|\; \text{Limit of form } \frac{(-)}{(+)(-)} \right. \\
\lim_{x\to -\frac{1}{2}^-} \frac{x^2-1}{2x^2-3x-2} &= \lim_{x\to -\frac{1}{2}^-} \frac{x^2-1}{2(x-2)\left(x+\frac{1}{2}\right)} = -\infty && \left|\; \text{Limit of form } \frac{(-)}{(-)(-)} \right.
\end{aligned}$$

Horizontal asymptotes. A function $f(x)$ has a horizontal asymptote if $\lim\limits_{x\to\pm\infty} f(x)$ exists. If that limit exists, and is some number, say, N, then $y=N$ is the equation of the corresponding asymptote.

We have that

$$\begin{aligned}
\lim_{x\to\infty} \frac{x^2-1}{2x^2-3x-2} &= \lim_{x\to\infty} \frac{\left(x^2-1\right)\frac{1}{x^2}}{\left(2x^2-3x-2\right)\frac{1}{x^2}} && \left|\; \text{Divide by highest term in den.} \right. \\
&= \lim_{x\to\infty} \frac{1-\frac{1}{x^2}}{2-\frac{3}{x}-\frac{2}{x^2}} \\
&= \frac{\lim\limits_{x\to\infty} 1 - \lim\limits_{x\to\infty} \frac{1}{x^2}}{\lim\limits_{x\to\infty} 2 - \lim\limits_{x\to\infty} \frac{3}{x} - \lim\limits_{x\to\infty} \frac{2}{x^2}} && \left|\; \text{Step may be skipped} \right. \\
&= \frac{1-0}{2-0-0} \\
&= \frac{1}{2}
\end{aligned}$$

A similar computation shows that

$$\lim_{x\to-\infty} \frac{x^2-1}{2x^2-3x-2} = \frac{1}{2}$$

Therefore $y=\frac{1}{2}$ is the only horizontal asymptote, valid in both directions ($x\to\pm\infty$).

A computer generated graph confirms our computations.

Solution. 22.f

Vertical asymptotes. The function is rational, and therefore has a finite limit (and therefore no vertical asymptote) at every point it its domain. The function is not defined for $x^2-2x-3=0$, which has two solutions, $x=-1$ and $x=3$. These are precisely the vertical asymptotes: indeed,

$$\begin{array}{rcll} \displaystyle\lim_{x\to -1^+} \frac{-5x^2-3x+5}{x^2-2x-3} &=& \displaystyle\lim_{x\to -1^+} \frac{-5x^2-3x+5}{(x+1)(x-3)} = -\infty & \Big| \text{ Limit of form } \frac{(+)}{(+)(-)} \\ \displaystyle\lim_{x\to -1^-} \frac{-5x^2-3x+5}{x^2-2x-3} &=& \displaystyle\lim_{x\to -1^-} \frac{-5x^2-3x+5}{(x+1)(x-3)} = \infty & \Big| \text{ Limit of form } \frac{(+)}{(-)(-)} \end{array}$$

and

$$\begin{array}{rcll} \displaystyle\lim_{x\to 3^+} \frac{-5x^2-3x+5}{x^2-2x-3} &=& \displaystyle\lim_{x\to 3^+} \frac{-5x^2-3x+5}{(x+1)(x-3)} = -\infty & \Big| \text{ Limit of form } \frac{(-)}{(+)(+)} \\ \displaystyle\lim_{x\to 3^-} \frac{-5x^2-3x+5}{x^2-2x-3} &=& \displaystyle\lim_{x\to 3^-} \frac{-5x^2-3x+5}{(x+1)(x-3)} = \infty & \Big| \text{ Limit of form } \frac{(-)}{(+)(-)} \end{array}$$

Horizontal asymptotes.

$$\begin{array}{rcll} \displaystyle\lim_{x\to\pm\infty} \frac{-5x^2-3x+5}{x^2-2x-3} &=& \displaystyle\lim_{x\to\pm\infty} \frac{\left(-5x^2-3x+5\right)\frac{1}{x^2}}{\left(x^2-2x-3\right)\frac{1}{x^2}} & \Big| \text{ Divide by highest term in den.} \\ &=& \displaystyle\lim_{x\to\pm\infty} \frac{-5-\frac{3}{x}+\frac{5}{x^2}}{1-\frac{2}{x}-\frac{3}{x^2}} & \\ &=& \dfrac{-\displaystyle\lim_{x\to\pm\infty} 5-\lim_{x\to\pm\infty}\frac{3}{x}+\lim_{x\to\pm\infty}\frac{5}{x^2}}{\displaystyle\lim_{x\to\pm\infty} 1-\lim_{x\to\pm\infty}\frac{2}{x}-\lim_{x\to\pm\infty}\frac{3}{x^2}} & \Big| \text{ Step may be skipped} \\ &=& \dfrac{-5-0+0}{1-0-0} & \\ &=& -5. & \end{array}$$

Therefore $y=-5$ is the only horizontal asymptote, valid in both directions ($x\to\pm\infty$).

A computer generated graph confirms our computations.

Solution. 22.k

Vertical asymptotes. A function $f(x)$ has a vertical asymptote at $x=a$ if $\lim_{x\to a} f(x)=\pm\infty$.

The function is algebraic, and therefore has a finite limit at every point it is defined (i.e., no asymptote). Therefore the function can have vertical asymptotes only for those x for which $f(x)$ is not defined. The function is not defined for

$$\begin{array}{rcll} \sqrt{x^2+3}-2x &=& 0 & \\ \sqrt{x^2+3} &=& 2x & \Big| \text{ square both sides may introduce extraneous solutions} \\ x^2+3 &=& 4x^2 & \\ 3x^2-3 &=& 0 & \\ 3(x-1)(x+1) &=& 0 & \\ x=1 & \text{or} & \cancel{x=-1} & \\ && x=-1 \text{ is extraneous:} & \\ && \sqrt{(-1)^2+3}-(-1)2=4\neq 0 & \end{array}$$

$x=-1$ is indeed a vertical asymptote:

$$\lim_{x\to 1^+} \frac{x}{\sqrt{x^2+3}-2x}=\infty \qquad \lim_{x\to 1^-} \frac{x}{\sqrt{x^2+3}-2x}=-\infty.$$

Horizontal asymptotes.

$$
\begin{aligned}
\lim_{x\to-\infty}\frac{x}{\sqrt{x^2+3}-2x} &= \lim_{x\to-\infty}\frac{1}{\frac{\sqrt{x^2+3}}{x}-2} \\
&= \lim_{x\to-\infty}\frac{1}{-\sqrt{\frac{x^2+3}{x^2}}-2} \qquad \left|\ \frac{1}{x}=-\sqrt{\frac{1}{x^2}} \text{ when } x<0\right. \\
&= \lim_{x\to-\infty}\frac{1}{-\sqrt{1+\frac{3}{x^2}}-2} \\
&= \frac{1}{-\sqrt{1+0}-2} \\
&= -\frac{1}{3}.
\end{aligned}
$$

$$
\begin{aligned}
\lim_{x\to-\infty}\frac{x}{\sqrt{x^2+3}-2x} &= \lim_{x\to\infty}\frac{1}{\frac{\sqrt{x^2+3}}{x}-2} \\
&= \lim_{x\to\infty}\frac{1}{\sqrt{\frac{x^2+3}{x^2}}-2} \qquad \left|\ \frac{1}{x}=\sqrt{\frac{1}{x^2}} \text{ when } x>0\right. \\
&= \lim_{x\to\infty}\frac{1}{\sqrt{1+\frac{3}{x^2}}-2} \\
&= \frac{1}{\sqrt{1+0}-2} \\
&= -1.
\end{aligned}
$$

Therefore $y=-\frac{1}{3}$ and $y=-1$ are the two horizontal asymptotes.
A computer generated graph confirms our computations.

23. Find the inverse function. You are asked to do the algebra only; you are not asked to determine the domain or range of the function or its inverse.

(a) $f(x)=3x^2+4x-7$, where $x\geq -\frac{2}{3}$.

answer: $f^{-1}(x) = -\frac{2}{3}+\frac{\sqrt{25+3x}}{3}, \quad x\geq -\frac{25}{3}$

(b) $f(x)=2x^2+3x-5$, where $x\geq -\frac{3}{4}$.

answer: $f^{-1}(x) = -\frac{3}{4}+\frac{\sqrt{49+8x}}{4}, \quad x\geq -\frac{49}{8}$

(c) $f(x)=\dfrac{2x+5}{x-4}$, where $x\neq 4$.

answer: $f^{-1}(x) = \frac{4x+5}{x-2}, \quad x\neq 2$

(d) $f(x)=\dfrac{3x+5}{2x-4}$, where $x\neq 2$.

answer: $f^{-1}(x) = \dfrac{4x+5}{2x-3}, \quad x\neq \dfrac{3}{2}$

(e) $f(x)=\dfrac{5x+6}{4x+5}$, where $x\neq -\frac{5}{4}$.

answer: $f^{-1}(x) = \frac{-5x+6}{4x-5}, x\neq\frac{5}{4}$

(f) $f(x)=\dfrac{2x-3}{-3x+4}$, where $x\neq \frac{4}{3}$..

answer: $f^{-1}(x) = \frac{4x+3}{3x+2}, x\neq -\frac{2}{3}$

Solution. 23.d This is a concise solution written in form suitable for test taking.

$$
\begin{aligned}
y &= \frac{3x+5}{2x-4} \\
y(2x-4) &= 3x+5 \\
2xy-4y &= 3x+5 \\
2xy-3x &= 4y+5 \\
x(2y-3) &= 4y+5 \\
x &= \frac{4y+5}{2y-3} \\
\text{Therefore} \quad f^{-1}(y) &= \frac{5+4y}{2y-3} \\
f^{-1}(x) &= \frac{5+4x}{2x-3}.
\end{aligned}
$$

Solution. 23.e. Set $f(x) = y$. Then

$$\begin{aligned} y &= \frac{5x+6}{4x+5} \\ y(4x+5) &= 5x+6 \\ x(4y-5) &= -5y+6 \\ x &= \frac{-5y+6}{4y-5}. \end{aligned}$$

Therefore the function $x = g(y) = \dfrac{-5y+6}{4y-5}$ is the inverse of $f(x)$. We write $g = f^{-1}$. The function $g = f^{-1}$ is defined for $y \neq \dfrac{5}{4}$. For our final answer we relabel the argument of g to x:

$$g(x) = f^{-1}(x) = \frac{-5x+6}{4x-5} \quad .$$

Let us check our work. In order for f and g to be inverses, we need that $g(f(x))$ be equal to x.

$$g(f(x)) = \frac{-5f(x)+6}{4f(x)-5} = \frac{-5\frac{(5x+6)}{4x+5}+6}{4\frac{(5x+6)}{4x+5}-5} = \frac{-5(5x+6)+6(4x+5)}{4(5x+6)-5(4x+5)} = \frac{-x}{-1} = x \quad ,$$

as expected.

24. Find the inverse function and its domain.

(a) $y = \ln(x+3)$.

answer: $f^{-1}(x) = e^{x} - 3$

(b) $y = 4\ln(x-3) - 4$.

answer: $f^{-1}(x) = e^{\frac{x+4}{4}x} + 3$

(c) $y = 2\ln(-2x+4) + 1$

answer: $f^{-1}(x) = -\frac{1}{2}e^{\frac{1}{2}x-\frac{1}{2}} + 2$

(d) $f(x) = e^{x^3}$.

answer: $f^{-1}(x) = \sqrt[3]{\ln x}, \quad x > 0$

(e) $y = (\ln x)^2$, $x \geq 1$.

answer: $f^{-1}(x) = e^{\sqrt{x}}, \quad x \geq 0$

(f) $y = \dfrac{e^x}{1+2e^x}$.

answer: $f^{-1}(x) = \ln\left(\frac{x}{1-2x}\right), \quad x \in \left(0, \frac{1}{2}\right)$

(g) $f(x) = 2^{2x} + 2^x - 2$.

answer: $f^{-1}(x) = \log_2 \frac{-1+\sqrt{9+4x}}{2}, \quad x \geq -2$

Solution. 24.a

$$\begin{aligned} y &= \ln(x+3) \\ e^y &= e^{\ln(x+3)} \\ e^y &= x+3 \\ e^y - 3 &= x \\ \text{Therefore} \quad f^{-1}(y) &= e^y - 3. \end{aligned}$$

The domain of e^y is all real numbers, so the domain of f^{-1} is all real numbers.

Solution. 24.b

$$\begin{aligned} 4\ln(x-3) - 4 &= y \\ 4\ln(x-3) &= y+4 \\ \ln(x-3) &= \frac{y+4}{4} && \Big| \text{ exponentiate} \\ e^{\ln(x-3)} &= e^{\frac{y+4}{4}} \\ x-3 &= e^{\frac{y+4}{4}} \\ f^{-1}(y) = x &= e^{\frac{y+4}{4}} + 3 \\ f^{-1}(x) &= e^{\frac{x+4}{4}} + 3 && \Big| \text{ relabel.} \end{aligned}$$

The domain of f^{-1} is all real numbers (no restrictions on the domain).

Solution. 24.e

$$\begin{array}{rcl|l} y &=& (\ln x)^2 & \text{take } \sqrt{\ } \text{ on both sides, } y \geq 0 \\ \sqrt{y} &=& \ln x & \text{exponentiate} \\ e^{\sqrt{y}} &=& e^{\ln x} = x & \\ f^{-1}(y) &=& e^{\sqrt{y}} & \\ f^{-1}(x) &=& e^{\sqrt{x}} & \end{array}$$

Solution. 24.f

$$\begin{aligned} y &= \frac{e^x}{1+2e^x} \\ y(1+2e^x) &= e^x \\ y &= e^x(1-2y) \\ \frac{y}{1-2y} &= e^x \\ \ln\frac{y}{1-2y} &= \ln e^x \\ \ln\frac{y}{1-2y} &= x \\ \text{Therefore} \quad f^{-1}(y) &= \ln\frac{y}{1-2y}. \end{aligned}$$

The natural logarithm function is only defined for positive input values. Therefore the domain is the set of all y for which

$$\frac{y}{1-2y} > 0.$$

This inequality holds if the numerator and denominator are both positive or both negative. This happens if either

(a) $y > 0$ and $y < \frac{1}{2}$, or

(b) $y < 0$ and $y > \frac{1}{2}$.

The latter option is impossible, so the domain is $\{y \in \mathbb{R} \mid 0 < y < \frac{1}{2}\}$.

25. Find the exact value of each expression.

(a) $\log_5 125$.

answer: 3

(b) $\log_3 \dfrac{1}{27}$.

answer: -3

(c) $\ln\left(\dfrac{1}{e}\right)$.

answer: -1

(d) $\log_{10}\sqrt{10}$.

answer: $\frac{1}{2}$

(e) $e^{\ln 4.5}$.

answer: 4.5

(f) $\log_{10} 0.0001$.

answer: 4

(g) $\log_{1.5} 2.25$.

answer: 2

(h) $\log_5 4 - \log_5 500$.

answer: -3

(i) $\log_2 6 - \log_2 15 + \log_2 20$.

answer: 3

(j) $\log_3 100 - \log_3 18 - \log_3 50$.

answer: -2

(k) $e^{-2\ln 5}$.

answer: $\frac{1}{25}$

(l) $\ln\left(\ln e^{e^{10}}\right)$.

answer: 10

26. Use the definition of a logarithm to evaluate each of the following without using a calculator.

(a) $\log_2 16$

(b) $\log_3\left(\dfrac{1}{9}\right)$

(c) $\log_{10} 1000$

(d) $\log_6 36^{-\frac{2}{3}}$

(e) $\log_2(8\sqrt{2})$

(f) $\log_7\left(\dfrac{49^x}{343^y}\right)$

Solution.

$$\begin{array}{rcl} \log_7\left(\frac{49^x}{343^y}\right) & = & \log_7 49^x - \log_7 343^y \\ & = & x\log_7 49 - y\log_7 343 \\ \text{But } 49 = 7^2 \text{ and } 343 = 7^3, \text{ therefore } \log_7\left(\frac{49^x}{343^y}\right) & = & 2x - 3y. \end{array}$$

27. Express each of the following as a single logarithm.

(a) $\ln 4 + \ln 6 - \ln 5$

(b) $2\ln 2 - 3\ln 3 + 4\ln 4$

Solution.

$$\begin{aligned} 2\ln 2 - 3\ln 3 + 4\ln 4 &= \ln 2^2 - \ln 3^3 + \ln 4^4 \\ &= \ln 4 - \ln 27 + \ln 256 \\ &= \ln\left(\frac{4}{27}\right) + \ln 256 \\ &= \ln\left(\frac{4\cdot 256}{27}\right) \\ &= \ln\left(\frac{1024}{27}\right). \end{aligned}$$

(c) $\ln 36 - 2\ln 3 - 3\ln 2$

28. Solve each equation for x. If available, use a calculator to give an ($\approx$) answer in decimal notation. If available, use a calculator to verify your approximate solutions.

(a) $e^{7-4x} = 7.$

answer: $\frac{7-\ln 7}{4} \approx 1.263522$

(b) $\ln(2x-9) = 2.$

answer: $\frac{e^2+9}{2} \approx 8.194528$

(c) $\ln(x^2-2) = 3.$

answer: $\pm\sqrt{e^3+2} \approx \pm 4.699525$

(d) $2^{x-3} = 5.$

answer: $\log_2 5 + 3 = \frac{\ln 5}{\ln 2} + 3 \approx 5.321928$

(e) $\ln x + \ln(x-1) = 1.$

answer: $\frac{1}{2}\left(1+\sqrt{1+4e}\right) \approx 2.223$

(f) $e^{2x+1} = t.$

answer: $\frac{\ln t - 1}{2}$

(g) $\log_2(mx) = c.$

answer: $\frac{2^c}{m}$

(h) $e - e^{-2x} = 1.$

answer: $-\frac{1}{2}\ln(e-1) \approx -0.271$

(i) $8(1+e^{-x})^{-1} = 3.$

answer: $-\ln\frac{5}{3} = \ln\frac{3}{5} \approx -0.510826$

(j) $\ln(\ln x) = 1.$

answer: $e^e \approx 15.154$

(k) $e^{e^x} = 10.$

answer: $\ln(\ln 10) \approx 0.834$

(l) $\ln(2x+1) = 3 - \ln x.$

answer: $\frac{-1+\sqrt{1+8e^3}}{4} \approx 2.92878$

(m) $e^{2x} - 4e^x + 3 = 0.$

answer: $x = \ln 3 \approx 1.098612$, $x = 0$

(n) $e^{4x} + 3e^{2x} - 4 = 0.$

answer: $x = 0$

(o) $e^{2x} - e^x - 6 = 0.$

answer: $x = \ln 3$

(p) $4^{3x} - 2^{3x+2} - 5 = 0.$

answer: $x = \frac{\log_2 5}{3}$

(q) $3\cdot 2^x + 2\left(\frac{1}{2}\right)^{x-1} - 7 = 0.$

answer: $x = 0$ or $2 - \log_2 3 = 2 - \frac{\ln 3}{\ln 2}$

Solution. 28.d

$$\begin{array}{rcl|l} 2^{x-3} & = & 5 & \text{take } \log_2 \\ x-3 & = & \log_2(5) & \text{add 3 to both sides} \\ x & = & \log_2(5)+3 & \text{answer is complete} \\ & = & \dfrac{\ln 5}{\ln 2}+3 & \text{optional step: convert to } \ln \\ & \approx & 5.321928095 & \text{calculator} \end{array}$$

Solution. 28.h

$$\begin{array}{rcll} e-e^{-2x} &=& 1 & \\ e^{-2x} &=& e-1 & \Big|\ \text{apply } \ln \\ \ln e^{-2x} &=& \ln(e-1) & \\ -2x &=& \ln(e-1) & \\ x &=& -\dfrac{1}{2}\ln(e-1) & \\ &\approx& -0.270662427 & \Big|\ \text{calculator} \end{array}$$

Solution. 28.e

$$\begin{array}{rcl} \ln x+\ln(x-1) &=& 1 \\ \ln\left(x^2-x\right) &=& 1 \\ e^{\ln(x^2-x)} &=& e^1 \\ x^2-x &=& e \\ x^2-x-e &=& 0 \\ \text{Quadratic formula:}\quad x &=& \dfrac{-(-1)\pm\sqrt{(-1)^2-4(1)(-e)}}{2(1)} \\ &=& \dfrac{1\pm\sqrt{1+4e}}{2}. \end{array}$$

However $\frac{1-\sqrt{1+4e}}{2}$ is negative, so $\ln\left(\frac{1-\sqrt{1+4e}}{2}\right)$ is undefined. Hence the only solution is $x=\frac{1+\sqrt{1+4e}}{2}\approx 2.2229$.

Solution. 28.p

$$\begin{array}{rcll} 4^{3x}-2^{3x+2}-5 &=& 0 & \\ 4^{3x}-4\cdot 2^{3x}-5 &=& 0 & \Bigg|\ \text{Set } \begin{array}{rcl} 2^{3x} &=& u \\ 4^{3x} &=& u^2 \end{array} \\ u^2-4u-5 &=& 0 & \\ (u-5)(u+1) &=& 0 & \\ u=5 &\text{or}& u=-1 & \\ 2^{3x}=5 && 2^{3x}=-1 & \\ 3x=\log_2(5) && \text{no real solution} & \\ x=\dfrac{\log_2 5}{3} &&& \\ \text{Calculator: } x\approx 0.773976 &&& \end{array}$$

Solution. 28.q

$$\begin{array}{rcll} 3\cdot 2^x+2\left(\dfrac{1}{2}\right)^{x-1}-7 &=& 0 & \\ 3\cdot 2^x+2\left(\dfrac{1}{2}\right)^{x}\left(\dfrac{1}{2}\right)^{-1}-7 &=& 0 & \\ 3\cdot 2^x+4\left(\dfrac{1}{2}\right)^{x}-7 &=& 0 & \Big|\ \text{Set } 2^x=u \\ 3u+\dfrac{4}{u}-7 &=& 0 & \Big|\ \text{Multiply by } u \\ 3u^2-7u+4 &=& 0 & \\ (u-1)(3u-4) &=& 0 & \\ u=1 &\text{or}& 3u-4=0 & \\ 2^x=1 && u=\dfrac{4}{3} & \\ x=0 && 2^x=\dfrac{4}{3} & \\ && x=\log_2\dfrac{4}{3}=\log_2 4-\log_2 3 & \\ && x=2-\log_2 3 & \\ \text{Calculator:} && x\approx 0.415037 & \end{array}$$

29. Compute the derivative.

(a) $f(x) = 2^{2015}$.

answer: 0

(b) $f(x) = \pi^{2015}$.

answer: 0

(c) $f(x) = 2 - \frac{2}{3}x$.

answer: $-\frac{2}{3}$

(d) $f(x) = \frac{3}{4}x^8$.

answer: $6x^7$

(e) $f(x) = x^3 - 4x + 6$.

answer: $-4 + 3x^2$

(f) $f(t) = \frac{1}{2}t^6 - 3t^4 + t$.

answer: $3t^5 - 12t^3 + 1$

(g) $g(x) = x^2(1 - 2x)$.

answer: $2x - 6x^2$

(h) $h(x) = (x - 2)(2x + 3)$.

answer: $4x - 1$

(i) $f(x) = 2x^{-\frac{3}{4}}$.

answer: $-\frac{3}{2}x^{-\frac{7}{4}}$

(j) $f(x) = cx^{-6}$.

answer: $-6cx^{-7}$

(k) $A(x) = -\frac{12}{x^5}$.

answer: $60x^{-6}$

Solution. 29.g Approach 1. Uncover the parenthesis, and then differentiate:

$\left(x^2(1-2x)\right)' = \left(x^2 - 2x^3\right)' = 2x - 6x^2$

Approach 2. Use first the product rule and then simplify:

$$\begin{array}{rcl}\left(x^2(1-2x)\right)' &=& (x^2)'(1-2x) + x^2(1-2x)' \\ &=& 2x(1-2x) + x^2(-2) \\ &=& 2x - 4x^2 - 2x^2 \\ &=& 2x - 6x^2.\end{array}$$

Of course, both approaches lead to the same answer.

30. (a) Given that $f(0) = 5$, $f'(0) = -1$, $g(0) = -4$, $g'(0) = 1$ and $h(x) = f(x)g(x)$, find the derivative $h'(0)$.

answer: $h'(0) = (-1)\cdot(-4) + 5\cdot 1 = 9$

(b) Given that $f(2) = -3$, $f'(2) = 2$, $g(2) = 5$, $g'(2) = 1$ and $h(x) = f(x)g(x)$, find the derivative $h'(2)$.

(c) Given that $f(0) = 5$, $f'(0) = -1$, $g(0) = -4$, $g'(0) = 1$ and $h(x) = \frac{f(x)}{g(x)}$, find the derivative $h'(0)$.

answer: $h'(0) = \frac{(-1)\cdot(-4) - 5\cdot 1}{(-4)^2} = -\frac{1}{16}$

(d) Given that $f(1) = 2$, $f'(1) = -1$, $g(1) = -3$, $g'(1) = 1$, $h(1) = 0$, $h'(1) = 1$ and $j(x) = f(x)g(x)h(x)$, find the derivative $j'(1)$.

answer: $j'(1) = (-1)\cdot(-3)\cdot 0 + 2\cdot 1\cdot 0 + 2\cdot(-3)\cdot 1 = -6$

Solution. 30.b

$$\begin{array}{rcll}h'(x) &=& (f(x)g(x))' = f'(x)g(x) + f(x)g'(x) & \big| \text{ product rule} \\ h'(2) &=& f'(2)g(2) + f(2)g'(2) = 2\cdot 5 + (-3)1 = 7. &\end{array}$$

31. Compute the derivative.

(a) $y = x^{\frac{5}{3}} - x^{\frac{2}{3}}$.

answer: $\frac{5}{3}x^{\frac{2}{3}} - 2/3x^{-\frac{1}{3}}$

(b) $f(x) = \sqrt{x} - x$.

answer: $-1 + \frac{1}{2}x^{-\frac{1}{2}}$

(c) $y = \sqrt{x}(x - 1)$.

answer: $\frac{3}{2}x^{\frac{1}{2}} - \frac{1}{2}x^{-\frac{1}{2}}$

(d) $f(x) = (2x + 1)^2$.

answer: $4 + 8x$

(e) $f(x) = 4\pi x^2$.

answer: $8\pi x$

(f) $y = \frac{x^2 + 4x + 3}{\sqrt{x}}$.

answer: $2x^{-\frac{1}{2}} + \frac{3}{2}x^{\frac{1}{2}} - \frac{3}{2}x^{-\frac{3}{2}}$

(g) $y = \frac{\sqrt{x} + x}{x^2}$.

answer: $-x^{-2} - \frac{3}{2}x^{-\frac{5}{2}}$

(h) $f(x) = \left(x + x^{-1}\right)^3$.

answer: $3x^2 + 3 - 3x^{-2} - 3x^{-4}$

(i) $f(x) = \sqrt{2}x + \sqrt{5x}$.

answer: $\sqrt{2} + \frac{\sqrt{5}}{2}x^{-\frac{1}{2}} = \sqrt{2} + \frac{\sqrt{5}}{2\sqrt{x}}$

(j) $y = \sqrt[5]{x} + 4\sqrt{x^5}$.

answer: $10x^{\frac{3}{2}} + \frac{1}{5}x^{-\frac{4}{5}}$

(k) $y = \left(\sqrt{x} + \dfrac{1}{\sqrt[3]{x}}\right)^2$.

answer: $1 + \frac{1}{3}x^{-\frac{5}{6}} - \frac{2}{3}x^{-\frac{5}{3}}$

(l) $f(x) = (1 + 2x^2)(x - x^2)$.

answer: $1 - 2x + 6x^2 - 8x^3$

(m) $f(x) = \dfrac{x^4 - 5x^3 + \sqrt{x}}{x^2}$.

answer: $-5 + 2x - \frac{3}{2}x^{-\frac{5}{2}}$

(n) $f(x) = (2x^3 + 3)(x^4 - 2x)$.

answer: $-6 - 4x^3 + 14x^6$

(o) $f(x) = (1 + x + x^2)(2 - x^4)$.

answer: $2 + 4x - 4x^3 - 5x^4 - 6x^5$

(p) $g(y) = \left(\dfrac{1}{y^2} - \dfrac{3}{y^4}\right)(y + 5y^3)$.

answer: $5 + 9y^{-4} + 14y^{-2}$

(q) $f(x) = (x^3 - 2x)(x^{-4} + x^{-2})$.

answer: $1 + 6x^{-4} + x^{-2}$

(r) $f(x) = \dfrac{1 + 2x}{3 - 4x}$.

answer: $10(3 - 4x)^{-2}$

Solution. 31.k

$$\begin{aligned}
\left(\left(\sqrt{x} + \frac{1}{\sqrt[3]{x}}\right)^2\right)' &= \left(\left(x^{\frac{1}{2}} + x^{-\frac{1}{3}}\right)^2\right)' \\
&= \left(\left(x^{\frac{1}{2}}\right)^2 + 2x^{\frac{1}{2}}x^{-\frac{1}{3}} + \left(x^{-\frac{1}{3}}\right)^2\right)' \\
&= \left(x + 2x^{\frac{1}{6}} + x^{-\frac{2}{3}}\right)' \\
&= 1 + 2\cdot\frac{1}{6}x^{\frac{1}{6}-1} + \left(-\frac{2}{3}\right)x^{-\frac{2}{3}-1} \\
&= 1 + \frac{1}{3}x^{-\frac{5}{6}} - \frac{2}{3}x^{-\frac{5}{3}}.
\end{aligned}$$

32. Compute the derivative (with respect to the implied variable).

(a) $f(x) = \dfrac{x - 3}{x + 3}$.

answer: $6(3 + x)^{-2}$

(b) $y = \dfrac{x^3}{1 - x^2}$.

answer: $\frac{3x^2 - x^4}{(1 - x^2)^2}$

(c) $y = \dfrac{x + 1}{x^3 + x - 2}$.

answer: $\frac{-3 - 3x^2 - 2x^3}{(-2 + x + x^3)^2}$

(d) $y = \dfrac{x - 1}{x^3 + x - 2}$.

answer: $\frac{-2x - 1}{\left(x^2 + x + 2\right)^2}$

(e) $f(x) = \dfrac{x + 1}{x^3 + 1}$.

answer: $\frac{-2x + 1}{(x^2 - x + 2)^2}$

(f) $y = \dfrac{x^3 - 2x\sqrt{x}}{x}$.

answer: $2x - x^{-\frac{1}{2}}$

(g) $y = \dfrac{t}{(t - 1)^2}$.

answer: $-\frac{t + 1}{(t - 1)^3}$

(h) $y = \dfrac{t^2 + 2}{t^4 - 3t^2 + 1}$.

answer: $\frac{14t - 8t^3 - 2t^5}{(1 - 3t^2 + t^4)^2}$

(i) $g(t) = \dfrac{t - \sqrt{t}}{t^{\frac{1}{3}}}$.

answer: $-\frac{1}{6}t^{-\frac{5}{6}} + \frac{2}{3}t^{-\frac{1}{3}}$

(j) $y = ax^2 + bx + c$.

answer: $b + 2ax$

(k) $y = A + \dfrac{B}{x} + \dfrac{C}{x^2}$.

answer: $\frac{-Bx - 2C}{x^3}$

(l) $f(t) = \dfrac{2t}{2 + \sqrt{t}}$.

answer: $\frac{4 + t^{\frac{1}{2}}}{\left(2 + t^{\frac{1}{2}}\right)^2}$

(m) $y = \dfrac{cx}{1 + cx}$.

answer: $c(1 + cx)^{-2}$

(n) $y = \sqrt[3]{t}(t^2 + t + t^{-1})$.

answer: $-\frac{2}{3}t^{-\frac{5}{3}} + \frac{4}{3}t^{\frac{1}{3}} + \frac{7}{3}t^{\frac{4}{3}}$

(o) $y = \dfrac{u^6 - 2u^3 + 5}{u^2}$.

answer: $-2 - 10u^{-3} + 4u^3$

(p) $f(x) = \dfrac{ax + b}{cx + d}$.

answer: $\frac{ad - bc}{(d + cx)^2}$

(q) $f(x) = \dfrac{1 + x}{1 + \frac{2}{x}}$.

answer: $\frac{x^2 + 4x + 2}{(2 + x)^2}$

(r) $f(x) = \dfrac{1 + x}{1 + \frac{3}{x}}$.

answer: $\frac{x^2 + 6x + 3}{(3 + x)^2}$

(s) $f(x) = \dfrac{x}{x + \frac{c}{x}}$.

answer: $\frac{2xc}{(c + x^2)^2}$

Solution. 32.g This can be differentiated more efficiently using the chain rule, however let us show how the problem can be solved directly using the quotient rule.

$$
\begin{aligned}
\left(\frac{t}{(t-1)^2}\right)' &= \frac{(t)'(t-1)^2 - t\left((t-1)^2\right)'}{(t-1)^4}\\
&= \frac{(t-1)^2 - t\left(t^2-2t+1\right)'}{(t-1)^4}\\
&= \frac{(t-1)^2 - t\,(2t-2)}{(t-1)^4}\\
&= \frac{\cancel{(t-1)}\left((t-1)-2t\right)}{(t-1)^{\cancel{4}\,3}}\\
&= \frac{-t-1}{(t-1)^3}\\
&= -\frac{t+1}{(t-1)^3}
\end{aligned}
$$

Solution. 32.e

$$
\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{x+1}{x^3+1}\right) &= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\cancel{x+1}}{\cancel{(x+1)}(x^2-x+1)}\right)\\
&= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{x^2-x+1}\right)
\end{aligned}
$$

Variant I: use quotient rule.

$$
\begin{aligned}
&= \frac{\frac{\mathrm{d}}{\mathrm{d}x}(1)\cdot(x^2-x+1) - 1\cdot\frac{\mathrm{d}}{\mathrm{d}x}\left(x^2-x+1\right)}{\left(x^2-x+1\right)^2}\\
&= \frac{-2x+1}{\left(x^2-x+1\right)^2}
\end{aligned}
$$

Variant I: use chain rule.

$$
\begin{aligned}
&= \frac{\mathrm{d}}{\mathrm{d}x}\left(\left(x^2-x+1\right)^{-1}\right)\\
&= -(x^2-x+1)^{-2}\frac{\mathrm{d}}{\mathrm{d}x}(x^2-x+1)\\
&= -(x^2-x+1)^{-2}(2x-1)\\
&= \frac{-2x+1}{\left(x^2-x+1\right)^2}.
\end{aligned}
$$

33. Compute the derivative.

(a) $f(x) = 2x^3 - 3\cos x$.

answer: $6x^2 + 3\sin x$

(b) $f(x) = \sqrt{x}\cos x$.

answer: $-x^{\frac{1}{2}}\sin x + \frac{1}{2}x^{-\frac{1}{2}}\cos x$

(c) $f(x) = \sin x + \frac{1}{3}\cot x$.

answer: $\frac{-\frac{1}{3}+\cos x\sin^2 x}{\sin^2 x} = \cos x - \frac{1}{3}\csc^2 x$

(d) $y = 2\sec x - \csc x$.

answer: $\frac{\cos^3 x + 2\sin^3 x}{(\cos x\sin x)^2}$

(e) $y = \dfrac{1+\sin^2\theta}{\cos^3\theta}$.

answer: $y' = \frac{2\sin\theta\cos^2\theta + 3\sin^3\theta + 3\sin\theta}{\cos^4\theta}$

(f) $g(t) = 4\sec t + \tan t - \csc t + 3\cot t$.

answer: $4\sec t\tan t + \sec^2 t + \csc t\cot t - 3\csc^2 t$

(g) $y = c\cos t + t^2\sin t$.

answer: $-c\sin t + 2t\sin t + t^2\cos t$

(h) $y = u(a\cos u + b\cot u)$.

answer: $\frac{-au\sin^3 u + a\cos u\sin^2 u - bu + b\cos u\sin u}{\sin^2 u}$

(i) $y = \dfrac{x}{2-\tan x}$.

answer: $\frac{x - \cos x\sin x + 2\cos^2 x}{(2\cos x - \sin x)^2}$

(j) $y = \sin\theta\cos\theta$.

answer: $\cos(2\theta) = \cos^2\theta - \sin^2\theta$

(k) $f(\theta) = \dfrac{\sec\theta}{1+\sec\theta}$.

answer: $\frac{\sin\theta}{(1+\cos\theta)^2}$

(l) $y = \dfrac{\cos x}{1-\sin x}$.

answer: $\frac{1}{1-\sin x}$

(m) $y = \dfrac{t\sin t}{1+t}$.

answer: $\frac{\sin t + t\cos t + t^2\cos t}{(1+t)^2}$

(n) $y = \dfrac{1-\sec x}{\tan x}$.

answer: $\frac{\cos x - 1}{\sin^2 x}$

(o) $h(\theta) = \theta \csc\theta - \cot\theta$.

answer: $\frac{1 + \sin\theta - \theta\cos\theta}{\sin^2\theta}$

(p) $y = x^2 \sin x \tan x$.

answer: $\frac{2x\cos x \sin^2 x + 2x^2 \sin x \cos^2 x + x^2 \sin^3(x)}{\cos^2 x}$

34. Differentiate.

(a) $\tan x$.

answer: $\sec^2 x$

(b) $\cot x$.

answer: $-\csc^2 x$

(c) $\sec x$.

answer: $\sec x \tan x = \frac{\sin x}{\cos^2 x}$

(d) $\csc x$.

answer: $-\csc x \cot x = -\frac{\cos x}{\sin^2 x}$

(e) $\sec x \tan x$.

answer: $\sec x \tan^2 x + \sec^3 x$

(f) $\sec x + \tan x$.

answer: $\sec x(\tan x + \sec x)$

(g) $\sec^2 x$.

answer: $2\tan x \sec^2 x$

(h) $\csc^2 x$.

answer: $-2\cot x \csc^2 x$

(i) $f(x) = (\sec x)e^x$.

answer: $\sec x \tan x e^x + \sec x e^x = (\tan x + 1)(\sec x)e^x$

(j) $f(x) = (\tan x)e^x$.

answer: $\sec^2 x e^x + \tan x e^x$

(k) $\dfrac{\sin x}{x}$.

answer: $\frac{x\cos x - \sin x}{x^2}$

(l) $\dfrac{\sin x}{e^x}$.

answer: $\frac{\cos x - \sin x}{e^x}$

(m) $x(\cos x)e^x$.

answer: $e^x(x\cos x - x\sin x + \cos x)$

(n) $\dfrac{e^x}{\tan x}$.

answer: $e^x\left(\cot x - \csc^2 x\right)$

(o) $\dfrac{e^x}{\sec x} + \sec x$.

answer: $e^x(\cos x - \sin x) + \sec x \tan x$

Solution. 34i

$$\begin{array}{rcll} \dfrac{\mathrm{d}}{\mathrm{d}x}\left((\sec x)e^x\right) & = & \left(\dfrac{\mathrm{d}}{\mathrm{d}x}(\sec x)\right)e^x + (\sec x)\dfrac{\mathrm{d}}{\mathrm{d}x}\left(e^x\right) & \Big| \text{ product rule} \\ & = & \sec x \tan x e^x + \sec x e^x & \\ & = & (\tan x + 1)\sec x e^x & \end{array}$$

Solution. 34j

$$\begin{array}{rcll} \dfrac{\mathrm{d}}{\mathrm{d}x}\left((\tan x)e^x\right) & = & \left(\dfrac{\mathrm{d}}{\mathrm{d}x}(\tan x)\right)e^x + (\tan x)\dfrac{\mathrm{d}}{\mathrm{d}x}\left(e^x\right) & \Big| \text{ product rule} \\ & = & (\sec^2 x)e^x + (\tan x)e^x & \\ & = & (\sec^2 x + \tan x)e^x. & \end{array}$$

35. Compute the derivative using the chain rule.

(a) $f(x) = \sqrt{1 + x^2}$

answer: $x(x^2+1)^{-\frac{1}{2}}$

(b) $f(x) = \sqrt{3x^2 - x + 2}$.

answer: $\frac{6x-1}{2\sqrt{3x^2-x+2}}$

(c) $f(x) = \dfrac{x}{\sqrt{1 + \frac{2}{x^2}}}$.

answer: $\frac{\pm x^2}{\sqrt{x^2+2}}$

(d) $f(x) = \sqrt{1 - \sqrt{x}}$.

answer: $-\frac{1}{4}x^{-\frac{1}{2}}\left(-\sqrt{x}+1\right)^{-\frac{1}{2}}$

(e) $y = (\cos x)^{\frac{1}{2}}$

answer: $\frac{1}{2}x^{-\frac{1}{2}}\cos\left(\sqrt{x}\right) = \frac{\cos(\sqrt{x})}{2\sqrt{x}}$

(f) $f(x) = \sin^3 x$.

answer: $3\cos x \sin^2 x$

(g) $y = (1 + \cos x)^2$.

answer: $-2\cos x \sin x - 2\sin x = -\sin(2x) - 2\sin x$

(h) $f(x) = \dfrac{1}{\sin^3 x}$.

answer: $-\frac{3\cos x}{\sin^4 x}$

(i) $f(x) = \sqrt[3]{4 + 3\tan x}$.

answer: $(4 + 3\tan x)^{-\frac{2}{3}}\sec^2 x$

(j) $f(x) = (\cos x + 3\sin x)^4$.

answer: $4(\cos x + 3\sin x)^3(3\cos x - \sin x)$

(k) $y = \sin\left(\sqrt{x}\right)$

answer: $\frac{\cos\sqrt{x}}{2\sqrt{x}}$

(l) $y = \cos(4x)$

answer: $-4\sin(4x)$

(m) $\sec^2(3x^2)$.

answer: $12\dfrac{x\sin(3x^2)}{\left(\cos(3x^2)\right)^3}$

(n) $\csc^2(3x^2)$.

answer: $-12\dfrac{x\cos(3x^2)}{\left(\sin(3x^2)\right)^3}$

(o) e^{2x}.

answer: $2e^{2x}$

(p) e^{-x^2}

answer: $-2xe^{-x^2}$

(q) $e^{\sqrt{x}}$

answer: $\frac{1}{2\sqrt{x}}e^{\sqrt{x}}$

(r) $f(x) = e^{-\frac{1}{x}}$.

answer: $\dfrac{e^{-\frac{1}{x}}}{x^2}$

(s) 5^x.

answer: $(\ln 5)5^x$

(t) e^{2^x}.

answer: $e^{2^x}2^x(\ln 2)$

(u) 2^{3^x}.

answer: $2^{3^x}3^x(\ln 2)(\ln 3)$

(v) 3^{2^x}.

answer: $3^{2^x}2^x(\ln 2)(\ln 3)$

(w) $y = \sqrt{\sec(4x)}$

answer: $2\sqrt{\sec(4x)}\tan(4x) = 2(\sec(4x))^{\frac{3}{2}}\sin(4x)$.

(x) $y = x^2\tan(5x)$

answer: $2x\tan(5x) - 5x^2\sec^2(5x)$

(y) $y = \dfrac{1+\sin\left(x^2\right)}{1+\cos\left(x^2\right)}$.

answer: $\dfrac{2x\left(1+\cos\left(x^2\right)+\sin\left(x^2\right)\right)}{\left(1+\cos\left(x^2\right)\right)^2}$

Solution. 35.b

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{3x^2-x+2}\right) = \frac{(3x^2-x+2)'}{2\sqrt{3x^2-x+2}} = \frac{6x-1}{2\sqrt{3x^2-x+2}}.$$

Solution. 35.c

$$\begin{aligned}\left(\frac{x}{\sqrt{1+\frac{2}{x^2}}}\right)' &= \frac{\sqrt{1+\frac{2}{x^2}} - x\left(\sqrt{1+\frac{2}{x^2}}\right)'}{1+\frac{2}{x^2}} = \frac{\sqrt{1+\frac{2}{x^2}} - x\frac{\frac{1}{2}}{\sqrt{1+\frac{2}{x^2}}}\left(\frac{2}{x^2}\right)'}{1+\frac{2}{x^2}}\\ &= \frac{\sqrt{1+\frac{2}{x^2}} + \frac{2}{x^2\sqrt{1+\frac{2}{x^2}}}}{1+\frac{2}{x^2}} = \frac{x^2\left(1+\frac{2}{x^2}\right)+2}{x^2\left(1+\frac{2}{x^2}\right)^{\frac{3}{2}}} = \frac{x^2+4}{x^2\left(1+\frac{2}{x^2}\right)^{\frac{3}{2}}}\end{aligned}$$

Please note that this problem can be solved also by applying the transformation

$$\frac{x}{\sqrt{1+\frac{2}{x^2}}} = \frac{x}{\sqrt{\frac{x^2+2}{x^2}}} = \frac{x}{\frac{1}{\pm x}\sqrt{x^2+2}} = \frac{\pm x^2}{\sqrt{x^2+2}}$$

before differentiating, however one must not forget the $\pm$ sign arising from $\sqrt{x^2} = \pm x$. Our original approach resulted in more algebra, but did not have the disadvantage of dealing with the $\pm$ sign.

Solution. 35.d

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{1-\sqrt{x}}\right) &= \frac{\mathrm{d}}{\mathrm{d}x}\left(\left(1-x^{\frac{1}{2}}\right)^{\frac{1}{2}}\right) && \text{chain rule}\\ &= \frac{1}{2}\left(1-x^{\frac{1}{2}}\right)^{-\frac{1}{2}}\frac{\mathrm{d}}{\mathrm{d}x}\left(1-x^{\frac{1}{2}}\right)\\ &= -\frac{1}{4}x^{-\frac{1}{2}}\left(1-x^{\frac{1}{2}}\right)^{-\frac{1}{2}}\end{aligned}$$

Solution. 35.e

$$\begin{aligned}\text{Let}\quad & u = \cos x.\\ \text{Then}\quad & y = u^{\frac{1}{2}}.\\ \text{Chain Rule:}\quad & \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}u}\frac{\mathrm{d}u}{\mathrm{d}x}\\ & = \left(\frac{1}{2}u^{-\frac{1}{2}}\right)(-\sin x)\\ & = -\frac{1}{2}\sin x(\cos x)^{-\frac{1}{2}}.\end{aligned}$$

Solution. 35.g

$$\begin{aligned}
\text{Let} \quad & u = 1 + \cos x. \\
\text{Then} \quad & y = u^2. \\
\text{Chain Rule:} \quad & \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}u}\frac{\mathrm{d}u}{\mathrm{d}x} \\
& \quad = (2u)(-\sin x) \\
& \quad = -2\sin x(1+\cos x) \\
& \quad = -2\sin x - 2\sin x\cos x \\
& \quad = -2\sin x - \sin(2x). \quad \text{(This last step is optional.)}
\end{aligned}$$

Solution. 35.k

$$\begin{aligned}
\text{Let} \quad & u = \sqrt{x}. \\
\text{Then} \quad & y = \sin u. \\
\text{Chain Rule:} \quad & \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}u}\frac{\mathrm{d}u}{\mathrm{d}x} \\
& \quad = (\cos u)\left(\frac{1}{2}u^{-\frac{1}{2}}\right) \\
& \quad = \frac{\cos\left(\sqrt{x}\right)}{2\sqrt{x}}.
\end{aligned}$$

Solution. 35.r

$$\begin{aligned}
\tfrac{\mathrm{d}}{\mathrm{d}x}\left(e^{-\frac{1}{x}}\right) &= e^{-\frac{1}{x}}\tfrac{\mathrm{d}}{\mathrm{d}x}\left(-\tfrac{1}{x}\right) && \text{chain rule} \\
&= -e^{-\frac{1}{x}}\frac{\mathrm{d}}{\mathrm{d}x}\left(x^{-1}\right) \\
&= x^{-2}e^{-\frac{1}{x}} \\
&= \frac{e^{-\frac{1}{x}}}{x^2}
\end{aligned}$$

Solution. 35.w

$$\begin{aligned}
\text{Chain Rule:} \quad & \frac{\mathrm{d}y}{\mathrm{d}x} = \left(\frac{1}{2}(\sec(4x))^{-\frac{1}{2}}\right)\frac{\mathrm{d}}{\mathrm{d}x}(\sec(4x)) \\
\text{Chain Rule:} \quad & \frac{\mathrm{d}y}{\mathrm{d}x} = \left(\frac{1}{2\sqrt{\sec(4x)}}\right)(\sec(4x)\tan(4x))\frac{\mathrm{d}}{\mathrm{d}x}(4x) \\
& \quad = \left(\frac{1}{2\sqrt{\sec(4x)}}\right)(\sec(4x)\tan(4x))(4) \\
& \quad = \frac{2\sec(4x)\tan(4x)}{\sqrt{\sec(4x)}}
\end{aligned}$$

There are many ways to simplify this answer, including both of the following.

$$\begin{aligned}
&= 2\sqrt{\sec(4x)}\tan(4x). \\
&= 2(\sec(4x))^{\frac{3}{2}}\sin(4x).
\end{aligned}$$

Solution. 35.x

$$\text{Product Rule:}\quad \frac{\mathrm{d}y}{\mathrm{d}x} = (x^2)\frac{\mathrm{d}}{\mathrm{d}x}(\tan(5x)) + (\tan(5x))\frac{\mathrm{d}}{\mathrm{d}x}(x^2)$$

Use the Chain Rule to differentiate $\tan(5x)$ in the first term.

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} &= (x^2)(-5\sec^2(5x) + (\tan(5x))(2x)\\ &= 2x\tan(5x) - 5x^2\sec^2(5x).\end{aligned}$$

Solution. 35.y

$$\text{Quotient Rule:}\quad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\left(1+\cos\left(x^2\right)\right)\frac{\mathrm{d}}{\mathrm{d}y}\left(1+\sin\left(x^2\right)\right) - \left(1+\sin\left(x^2\right)\right)\frac{\mathrm{d}}{\mathrm{d}x}\left(1+\cos\left(x^2\right)\right)}{(1+\cos(x^2))^2}$$

By the Chain Rule, $\frac{\mathrm{d}}{\mathrm{d}x}\left(1+\cos\left(x^2\right)\right) = -2x\sin\left(x^2\right)$ and $\frac{\mathrm{d}}{\mathrm{d}x}\left(1+\sin\left(x^2\right)\right) = 2x\cos\left(x^2\right)$.

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} &= \frac{\left(1+\cos\left(x^2\right)\right)\left(2x\cos\left(x^2\right)\right) - \left(1+\sin\left(x^2\right)\right)\left(-2x\sin\left(x^2\right)\right)}{(1+\cos(x^2))^2}\\ &= \frac{2x\cos\left(x^2\right) + 2x\cos^2\left(x^2\right) + 2x\sin\left(x^2\right) + 2x\sin^2\left(x^2\right)}{(1+\cos(x^2))^2}\\ &= \frac{2x\left(\cos^2\left(x^2\right) + \sin^2\left(x^2\right)\right) + 2x\left(\cos\left(x^2\right) + \sin\left(x^2\right)\right)}{(1+\cos(x^2))^2}\end{aligned}$$

By the Pythagorean Identity, $\cos^2\left(x^2\right) + \sin^2\left(x^2\right) = 1$.

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} &= \frac{2x + 2x\left(\cos\left(x^2\right) + \sin\left(x^2\right)\right)}{(1+\cos(x^2))^2}\\ &= \frac{2x\left(1+\cos\left(x^2\right) + \sin\left(x^2\right)\right)}{(1+\cos(x^2))^2}.\end{aligned}$$

36. Compute the derivative.

(a) $f(x) = (x^4 + 3x^2 - 2)^5$.

answer: $\left(30x + 20x^3\right)\left(-2 + 3x^2 + x^4\right)^4$

(b) $f(x) = (4x - x^2)^{100}$.

answer: $(-200x + 400)\left(4x - x^2\right)^{99}$

(c) $f(x) = (2x-3)^4(x^2+x+1)^5$.

answer: $\left(-7 - 12x + 28x^2\right)(-3+2x)^3\left(1 + x + x^2\right)^4$

(d) $f(x) = (x^2+1)^3(x^2+2)^6$.

answer: $\left(24x + 18x^3\right)\left(1+x^2\right)^2\left(2+x^2\right)^5$

(e) $f(x) = (3x-1)^4(2x+1)^{-3}$.

answer: $(3x-1)^3\frac{6x+18}{(2x+1)^4}$

(f) $f(x) = \dfrac{1}{1+x^2}$.

answer: $\frac{-2x}{(1+x^2)^2}$

(g) $f(x) = \left(\dfrac{x^2+1}{x^2-1}\right)^3$.

answer: $\frac{-12x}{(x^2-1)^2}\left(\frac{x^2+1}{x^2-1}\right)^2$

(h) $f(x) = (x+1)^{\frac{2}{3}}(2x^2-1)^3$.

answer: $\left(\frac{40}{3}x^2 + 12x - \frac{2}{3}\right)\left(2x^2-1\right)^2(x+1)^{-\frac{1}{3}}$

(i) $f(x) = \sqrt{1-2x}$.

answer: $-(1-2x)^{-\frac{1}{2}}$

(j) $f(x) = \sqrt{\dfrac{x^2+1}{x^2+4}}$.

answer: $\frac{3x}{(x^2+4)^2}\left(\frac{x^2+1}{x^2+4}\right)^{-\frac{1}{2}}$

(k) $f(x) = 3\cot(2x)$.

answer: $\frac{-6}{(\sin(2x))^2}$

(l) $f(x) = \dfrac{1}{(1+\sec x)^2}$.

answer: $\frac{-2\cos(x)\sin(x)}{(1+\cos(x))^3} = \frac{-\sin(2x)}{(1+\cos(x))^3}$

(m) $f(x) = \sqrt[3]{1+\tan x}$.

answer: $\frac{1}{3}(1+\tan x)^{-\frac{2}{3}}\sec^2 x$

(n) $f(x) = \cos(2+x^3)$.

answer: $-3x^2\sin\left(2+x^3\right)$

(o) $f(x) = \cos\left(\frac{1}{x}\right)\sin(x^2)$.

answer: $x^{-2}\sin\left(x^{-1}\right)\sin\left(x^2\right) + 2x\cos\left(x^{-1}\right)\cos\left(x^2\right)$

(p) $f(x) = x\sec(kx)$.

answer: $\frac{\cos(kx) + kx\sin(kx)}{(\cos(kx))^2}$

37. Differentiate.

(a) $f(x) = \sin(\tan(2x))$.

answer: $2\sec^2(2x)\cos(\tan 2x)$

(b) $f(x) = \sec^2(mx)$.

answer: $\frac{2m\sin(mx)}{(\cos(mx))^3}$

(c) $f(x) = \sec^2 x + \tan^2 x$.

answer: $\frac{4\sin x}{\cos^3 x}$

(d) $f(x) = x\sin\left(\frac{1}{x}\right)$.

answer: $-x^{-1}\cos\left(x^{-1}\right) + \sin\left(x^{-1}\right)$

(e) $f(x) = \left(\frac{1-\cos(2x)}{1+\cos(2x)}\right)^4$.

answer: $\frac{16\sin(2x)}{(\cos(2x)+1)^2}\left(\frac{-\cos(2x)+1}{\cos(2x)+1}\right)^3$

(f) $f(x) = \sqrt{\frac{x}{x^2+4}}$.

answer: $\frac{-\frac{1}{2}x^2+2}{(x^2+4)^2}\left(\frac{x}{x^2+4}\right)^{-\frac{1}{2}}$

(g) $f(t) = \cot^2(\sin t)$.

answer: $\frac{-2\cos t\cos(\sin t)}{\sin^3(\sin t)}$

(h) $f(x) = \left(ax + \sqrt{x^2+b^2}\right)^{-2}$.

answer: $\frac{-2x\left(x^2+b^2\right)^{-\frac{1}{2}} - 2a}{\left(\left(x^2+b^2\right)^{\frac{1}{2}} + ax\right)^3}$

(i) $f(x) = \left(x^2 + (1-3x)^5\right)^3$.

answer:
$\left(-45(-3x+1)^4 + 6x\right)\left((-3x+1)^5 + x^2\right)^2$
Using computer algebra:
$(-3645x^4 + 4860x^3 - 2430x^2 + 546x - 45)\left((-3x+1)^5 + x^2\right)^2$
Using computer algebra full expansion:
$-215233605x^{14} + 1004423490x^{13} - 2176250895x^{12}$
$+2903793624x^{11} - 2666357595x^{10} + 1782098820x^9$
$-893713176x^8 + 341444160x^7 - 99805041x^6$
$+22199676x^5 - 3697470x^4 + 447132x^3$
$-37125x^2 + 1896x - 45$

(j) $f(x) = \sin(\sin(\sin x))$.

answer: $\cos x\cos(\sin x)\cos(\sin(\sin x))$

(k) $f(x) = \sqrt{x + \sqrt{x}}$.

answer: $\left(\frac{1}{2} + \frac{1}{4}x^{-\frac{1}{2}}\right)\left(x^{\frac{1}{2}} + x\right)^{-\frac{1}{2}}$

(l) $f(x) = \sqrt{x + \sqrt{x + \sqrt{x}}}$.

answer: $\frac{1}{2}\left(\left(x^{\frac{1}{2}} + x\right)^{\frac{1}{2}} + x\right)^{-\frac{1}{2}}\left(\frac{1}{2}\left(x^{\frac{1}{2}} + x\right)^{-\frac{1}{2}}\left(\frac{1}{2}x^{-\frac{1}{2}} + 1\right) + 1\right)$

(m) $f(x) = (2r\sin(rx) + n)^p$.

answer: $pr(2r\sin(rx) + n)^{p-1}\cos(rx)$

(n) $f(x) = \cos^4(\sin^3 x)$.

answer: $-12\cos x\sin^2 x\sin(\sin^3 x)\cos^3(\sin^3 x)$

(o) $f(x) = \cos\sqrt{\sin(\tan(\pi x))}$.

answer: $\frac{-\frac{1}{2}\pi\cos(\tan(\pi x))\sin\left(\sqrt{\sin(\tan(\pi x))}\right)}{\sqrt{\sin(\tan(\pi x))}\cos^2(\pi x)}$

(p) $f(x) = \left(x + (x + \sin^2 x)^3\right)^4$.

answer: $4((\sin^2 x + x)^3 + x)^3(3(\sin^2 x + x)^2(2\sin x\cos x + 1) + 1)$

38. Compute the second derivative.

(a) $f(x) = \sin(-5x)$.

answer: $25\sin(5x)$

(b) $f(x) = \cot(2x)$.

answer: $8\cot(2x)\csc^2(2x) = \frac{8\cos(2x)}{\sin^3(2x)}$

(c) $f(x) = e^{-3x}$.

answer: $9e^{-3x}$

(d) $f(x) = e^{\frac{1}{x}}$.

answer: $2e^{x^{-1}}x^{-3} + e^{x^{-1}}x^{-4}$

(e) $f(x) = e^{\sqrt{x}}$.

answer: $e^{x^{\frac{1}{2}}}x^{-\frac{3}{2}} + \frac{1}{4}e^{x^{\frac{1}{2}}}x^{-1}$

(f) $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

answer: $\frac{-8\left(-e^{-x}+e^x\right)}{\left(e^{-x}+e^x\right)^3}$

(g) $f(x) = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$

answer: $\frac{1}{2}\left(-\frac{1}{(x+1)^2} + \frac{1}{(-x+1)^2}\right) = \frac{2x}{\left(1-x^2\right)^2}$

39. Compute the derivative.

(a) $\ln(4x)$

answer: $\frac{1}{x}$

(b) $\ln(-13x)$

answer: $\frac{1}{x}$

(c) $\log_2(5x)$

answer: $\frac{1}{\ln(2)x}$

(d) $\log_{10}(-3x)$

answer: $\frac{1}{\ln(10)x}$

(e) $x^6\ln(2x)$

answer: $6x^5\ln(2x) + x^5$

(f) $x^4\ln(2x)$

answer: $4x^3\ln(2x) + x^3$

(g) $\ln\left(x^4\right)$

answer: $\frac{4}{x}$

(h) $(\ln(x))^4$

answer: $\frac{4(\ln x)^3}{x}$

(i) $\ln(7x+1)$

answer: $\frac{7}{7x+1}$

(j) $\ln\left(-6x+2\right)$

answer: $\frac{3}{3x-1}$

(k) $\ln\left(\frac{3x-2}{-2x+3}\right)$

answer: $\frac{-5}{6x^2-13x+6} = -\frac{3}{(-3x+2)} - \frac{2}{(2x-3)}$

(l) $\ln\left(\frac{5x-4}{-x-5}\right)$

answer: $\frac{29}{5x^2+21x-20} = \frac{5}{5x-4} - \frac{1}{x+5}$

(m) $\ln\left(\frac{3x+1}{4x-5}\right)$.

answer: $\frac{3}{3x+1} - \frac{4}{4x-5} = \frac{-19}{12x^2-11x-5}$

(n) $\ln\left(\cot x\right)$

answer: $-\csc x \sec x$

(o) $\ln\left(\sec(2x)\right)$

answer: $2\tan(2x)$

(p) $f(x) = \ln(\sec x) + \ln(\cot x)$.

answer: $-\cot x$

Solution. 39.k

$$\begin{array}{rcll}\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\frac{3x-2}{-2x+3}\right)\right) &=& \frac{\mathrm{d}}{\mathrm{d}x}\left(\ln(3x-2) - \ln(-2x+3)\right) & \text{logarithm properties}\\ &=& \frac{(3x-2)'}{(3x-2)} - \frac{(-2x+3)'}{-2x+3} & \text{chain rule}\\ &=& \frac{3}{3x-2} - \frac{-2}{-2x+3} & \\ &=& \frac{3}{3x-2} - \frac{2}{2x-3} & \\ &=& \frac{-5}{6x^2-13x+6} & \text{combine fractions (optional).}\end{array}$$

Solution. 39.m

$$\begin{array}{rcll}\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\frac{3x+1}{4x-5}\right)\right) &=& \frac{\mathrm{d}}{\mathrm{d}x}\left(\ln(3x+1) - \ln(4x-5)\right) & \\ &=& \frac{(3x+1)'}{3x+1} - \frac{(4x-5)'}{4x-5} & \\ &=& \frac{3}{3x+1} - \frac{4}{4x-5} & \\ &=& \frac{-19}{12x^2-11x-5} & \text{step optional.}\end{array}$$

Solution. 39.p

$$\begin{array}{rcll}\frac{\mathrm{d}}{\mathrm{d}x}(\ln(\sec x) + \ln(\cot x)) &=& \frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\frac{1}{\cos x}\right) + \ln\left(\frac{\cos x}{\sin x}\right)\right) & \\ &=& \frac{\mathrm{d}}{\mathrm{d}x}\left(\cancel{-\ln(\cos x)} + \cancel{\ln(\cos x)} - \ln(\sin x)\right) & \\ &=& -\frac{\mathrm{d}}{\mathrm{d}x}(\ln(\sin x)) & \text{chain rule}\\ &=& -\frac{1}{\sin x}\frac{\mathrm{d}}{\mathrm{d}x}(\sin x) & \\ &=& -\frac{\cos x}{\sin x} & \\ &=& -\cot x & \end{array}$$

40. Differentiate.

(a) 10^{x^3}.

answer: $3(\ln 10)x^2(10)^{x^3}$

(b) $2^{\tan x}$.

answer: $(\ln 2)2^{\tan x}\sec^2 x$

(c) x^x.

answer: $x^x(\log(x) + 1)$

(d) x^{x^x}.

answer: $(\ln(x))^2 x^{x^x+x} + x^{x^x+x-1} + (\ln x)x^{x^x+x}$

(e) $(\sin x)^{\cos x}$.

answer: $\frac{-\ln(\sin x)(\sin x)^{\cos x+2} + (\sin x)^{\cos x}\cos^2 x}{\sin x}$

(f) $(\ln x)^{\ln x}$.

answer: $\ln(\ln(x))x^{-1}(\ln(x))^{\ln(x)} + x^{-1}(\ln(x))^{\ln(x)}$

41. Find the limit.

(a) $\lim_{x\to\infty}\left(1-\frac{2}{x}\right)^x$. answer: e^{-2}

(b) $\lim_{x\to 0}(1-x)^{\frac{1}{x}}$. answer: e^{-1}

(c) $\lim_{x\to\infty}\left(\frac{x}{x-5}\right)^x$. answer: e^5

(d) $\lim_{x\to\infty}\left(\frac{x}{x-2}\right)^{3x+2}$. answer: e^6

42.

(a) Find the linearization of $f(x)=\sqrt{x}$ at $a=100$ and use it to approximate $\sqrt{99.8}$.

answer: $L(x)=10+0.05(x-100)$. Therefore $\sqrt{99.8}\approx L(99.8)=9.99$.

(b) Find the linearization of $f(x)=\sqrt{8+x}$ at $a=1$ and use it to approximate $\sqrt{9.02}$.

answer: $f(x)\approx 3+\frac{1}{6}(x-1)=\frac{1}{6}x+\frac{17}{6}$. Therefore $\sqrt{9.02}\approx\frac{901}{300}\approx 3.003333$

(c) Find the linearization of $f(x)=\sqrt[3]{8+x}$ at $a=0$ and use it to approximate $\sqrt[3]{7.97}$.

answer: $\sqrt[3]{8+x}\approx\frac{1}{12}x+2$. Therefore $\sqrt[3]{7.97}\simeq\frac{799}{400}=1.9975$

(d) Find the linearization of $f(x)=\ln x$ at $a=1$ and use it to approximate $\ln 1.01$.

answer: $f(x)\approx f(1)+f'(1)(x-1)=x-1, \ln 1.01\approx 0.01$.

(e) Use a linear approximation to estimate $(1.001)^9$.

answer: $(1.001)^9\approx 1.009$.

(f) Use a linear approximation to estimate $(0.9999)^{2014}$.

answer: $(0.9999)^{2014}\approx 0.7986$.

Solution. 42.f Let $f(x)=x^{2014}$. We are looking to approximate $(0.9999)^{2014}=f(0.9999)$. As $f(1)=1^{2014}=1$ is easy to compute, is makes sense to use linear approximation at $a=1$ to approximate $(0.9999)^{2014}$. We have that

$$f'(x)=2014x^{2013} \quad .$$

Therefore the linear approximation of $f(x)=x^{2014}$ at $a=1$ is:

$$f(x)\approx f(1)+f'(1)(x-1)=1^{2014}+2014\cdot 1^{2013}(x-1)=1+2014(x-1)=2014x-2013 \quad .$$

Therefore

$$f(0.9999)\approx 2014\cdot 0.9999-2013=1\cdot 0.9999+2013(0.9999-1)=0.9999-2013\cdot 0.0001=0.9999-0.2013=0.7986$$

A computation with computer shows that $0.999^{2014}=0.817577\ldots$. While our approximation of 0.7986 is less than perfect, it is within the same order of magnitude. We study techniques for estimating errors in linear approximations later.

43. Express $\frac{\mathrm{d}y}{\mathrm{d}x}$ as a function of x and y by implicit differentiation. The answer key has not been proofread, use with caution.

(a) $x^3+y^3=1$. answer: $\frac{\mathrm{d}y}{\mathrm{d}x}=-\frac{x^2}{y^2}$

(b) $2\sqrt{x}+\sqrt{y}=3$. answer: $\frac{\mathrm{d}y}{\mathrm{d}x}=-2\sqrt{\frac{y}{x}}$

(c) $x^2+xy-y^2=4$. answer: $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{-2x-y}{x-2y}$

(d) $2x^3+x^2y-xy^3=2$. answer: $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{y^3-6x^2-2xy}{-3xy^2+x^2}$

(e) $x^4(x+y)=y^2(3x-y)$. answer: $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{-5x^4-4x^3y+3y^2}{x^4-6xy-3y^2}$

(f) $y^5+x^2y^3=1+x^4y$. answer: $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{4x^3y-2xy^3}{5y^4+3x^2y^2-x^4}$

(g) $y\cos x=x^2+y^2$. answer: $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{y\sin x+2x}{\cos x-2y}$

(h) $\cos(xy)=1+\sin y$. answer: $\frac{\mathrm{d}y}{\mathrm{d}x}=-\frac{y\sin(xy)}{\cos y+x\sin(xy)}$

(i) $4\cos x\sin y=1$. answer: $\frac{\mathrm{d}y}{\mathrm{d}x}=\tan x\tan y$

(j) $y\sin\left(x^2\right)=x\sin\left(y^2\right)$. answer: $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{-2xy\cos\left(x^2\right)+\sin\left(y^2\right)}{-2xy\cos\left(y^2\right)+\sin\left(x^2\right)}$

(k) $\tan\left(\frac{x}{y}\right)=x+y$. answer: $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{-y^2+y\sec^2\left(\frac{y}{x}\right)}{y^2+x\sec^2\left(\frac{x}{y}\right)}$

(l) $\sqrt{x+y}=1+x^2y^2$. answer: $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{-\frac{1}{2}(y+x)^{-\frac{1}{2}}+8xy^2}{\frac{1}{2}(y+x)^{-\frac{1}{2}}-8x^2y}$

(m) $\sqrt{xy} = 1 + x^2y$.

answer: $\frac{dy}{dx} = \frac{2xy - \frac{y}{2\sqrt{xy}}}{\frac{x}{2\sqrt{xy}} - x^2} = \frac{4xyx^2y + 4xy - y}{-2x^4y - 2x^2y + x}$

(n) $x \sin y + y \sin x = 1$.

answer: $\frac{dy}{dx} = \frac{-y \cos x - \sin y}{x \cos y + \sin x}$

(o) $y \cos x = 1 + \sin(xy)$.

answer: $\frac{dy}{dx} = \frac{\cos(xy)y + y \sin x}{-\cos(xy)x + \cos x}$

(p) $\tan(x - y) = \dfrac{y}{1 + x^2}$.

answer: $\frac{dy}{dx} = \frac{-(\sec(-y+x))^2 x^4 - 2(\sec(-y+x))^2 x^2 - (\sec(-y+x))^2 - 2yx}{-(\sec(-y+x))^2 x^4 - 2(\sec(-y+x))^2 x^2 - (\sec(-y+x))^2 - x^2 - 1}$

(q) $x^4(x + y) = y^2(3x - y)$.

answer: $\frac{dy}{dx} = \frac{-4x^3y + 3y^2 - 5x^4}{3y^2 + x^4 - 6yx}$

(r) $2x^3 + x^2y - xy^3 = 2$.

answer: $\frac{dy}{dx} = \frac{y^3 - 6x^2 - 2xy}{-3xy^2 + x^2}$

44. Verify that the coordinates of the given point satisfy the given equation. Use implicit differentiation to find an equation of the tangent line to the curve at the given point.

(a) $y \sin(2x) = x \cos(2y)$, $\left(\frac{\pi}{2}, \frac{\pi}{4}\right)$.

answer: $y = \frac{1}{2}x$

(b) $\sin(x + y) = 2x - 2y$, (π, π) .

answer: $\frac{1}{3}x + \frac{2}{3}\pi$

(c) $x^2 + xy + y^2 = 3$, $(1, -2)$ (ellipse).

answer: $y = -2$

(d) $x^2 + 2xy - y^2 + x = 2$, $(1, 2)$ (hyperbola).

answer: $y = \frac{7}{2}x - \frac{3}{2}$

(e) $y^3 + x^3 + 4xy = \dfrac{3}{4}$, $\left(-\frac{1}{2}, -\frac{1}{2}\right)$

answer: $y = -x - 1$

(f) $y^3 + x^3 + 4xy = -4$, $(1, -1)$

answer:

(g) $x^2 + y^2 = (2x^2 + 2y^2 - x)^2$, $(0, \frac{1}{2})$.

answer: $y = x + \frac{1}{2}$

(h) $x^{\frac{2}{3}} + y^{\frac{2}{3}} = 4$, $(-3\sqrt{3}, 1)$.

answer: $y = \frac{1}{\sqrt{3}}x + 4$

(i) $2(x^2 + y^2)^2 = 25(x^2 - y^2)$, $(3, 1)$.

answer: $y = -\frac{9}{13}x + \frac{40}{13}$

(j) $y^2(y^2 - 4) = x^2(x^2 - 5)$, $(0, -2)$.

answer: $y = -2$

(k) $x^{\frac{4}{3}} + y^{\frac{4}{3}} = 10$ at $(-3\sqrt{3}, 1)$.

answer: $y = \sqrt{3}x + 10$

(l) $x^2y^3 + x^3 - y^2 = 1$ at $(1, 1)$.

answer: $y = -5x + 6$

Solution. 44.a

First we verify that the point $(x, y) = \left(\dfrac{\pi}{2}, \dfrac{\pi}{4}\right)$ indeed satisfies the given equation:

$$\begin{array}{rcll} y\sin(2x)_{|x=\frac{\pi}{2}, y=\frac{\pi}{4}} = \dfrac{\pi}{4}\sin\pi & = & 0 & \Big| \text{ left hand side} \\ x\cos(2y)_{|x=\frac{\pi}{2}, y=\frac{\pi}{4}} = \dfrac{\pi}{2}\cos\left(\dfrac{\pi}{2}\right) & = & 0 & \Big| \text{ right hand side} \end{array}$$

so the two sides of the equation are equal (both to 0) when $x = \frac{\pi}{2}$ and $y = \frac{\pi}{4}$.

Since we are looking an equation of the tangent line, we need to find $\frac{\mathrm{d}y}{\mathrm{d}x}_{|x=\frac{\pi}{2}, y=\frac{\pi}{4}}$ - that is, the derivative of y at the point $x = \frac{\pi}{2}$, $y = \frac{\pi}{4}$. To do so we use implicit differentiation.

$$\begin{array}{rcll} y\sin(2x) & = & x\cos(2y) & \Big| \frac{\mathrm{d}}{\mathrm{d}x} \\ \dfrac{\mathrm{d}y}{\mathrm{d}x}\sin(2x) + y\dfrac{\mathrm{d}}{\mathrm{d}x}(\sin(2x)) & = & \cos(2y) + x\dfrac{\mathrm{d}}{\mathrm{d}x}(\cos(2y)) & \\ \dfrac{\mathrm{d}y}{\mathrm{d}x}\sin(2x) + 2y\cos(2x) & = & \cos(2y) - 2x\sin(2y)\dfrac{\mathrm{d}y}{\mathrm{d}x} & \\ \dfrac{\mathrm{d}y}{\mathrm{d}x}(\sin(2x) + 2x\sin(2y)) & = & \cos(2y) - 2y\cos(2x) & \Big| \text{Set } x = \frac{\pi}{2}, y = \frac{\pi}{4} \\ \dfrac{\mathrm{d}y}{\mathrm{d}x}_{|x=\frac{\pi}{2}, y=\frac{\pi}{4}}\left(\sin\pi + \pi\sin\left(\dfrac{\pi}{2}\right)\right) & = & \cos\left(\dfrac{\pi}{2}\right) - \dfrac{\pi}{2}\cos\pi & \\ \pi\dfrac{\mathrm{d}y}{\mathrm{d}x}_{|x=\frac{\pi}{2}, y=\frac{\pi}{4}} & = & -\frac{\pi}{2}\cos\pi & \\ \dfrac{\mathrm{d}y}{\mathrm{d}x}_{|x=\frac{\pi}{2}, y=\frac{\pi}{4}} & = & \dfrac{1}{2}. & \end{array}$$

Therefore the equation of the line through $x = \frac{\pi}{2}, y = \frac{\pi}{4}$ is

$$\begin{array}{rcl} y - \dfrac{\pi}{4} & = & \dfrac{1}{2}\left(x - \dfrac{\pi}{2}\right) \\ y & = & \dfrac{1}{2}x. \end{array}$$

Solution. 44.e

$$\begin{array}{rcll}
y^3+x^3+4xy &=& \dfrac{3}{4} & \Big| \text{ apply } \frac{\mathrm{d}}{\mathrm{d}x} \\
3y^2\dfrac{\mathrm{d}y}{\mathrm{d}x}+3x^2+4\left(y+x\dfrac{\mathrm{d}y}{\mathrm{d}x}\right) &=& 0 & \\
\dfrac{\mathrm{d}y}{\mathrm{d}x}\left(3y^2+4x\right) &=& -3x^2-4y & \\
\dfrac{\mathrm{d}y}{\mathrm{d}x} &=& \dfrac{-3x^2-4y}{3y^2+4x} & \Big| \text{ substitute } x=-\frac{1}{2}, y=-\frac{1}{2} \\
\dfrac{\mathrm{d}y}{\mathrm{d}x}_{|x=-\frac{1}{2},y=-\frac{1}{2}} &=& \dfrac{-3\left(-\frac{1}{2}\right)^2-4\left(-\frac{1}{2}\right)}{3\left(-\frac{1}{2}\right)^2+4\left(-\frac{1}{2}\right)} & \\
\dfrac{\mathrm{d}y}{\mathrm{d}x}_{|x=-\frac{1}{2},y=-\frac{1}{2}} &=& -1 &
\end{array}$$

Therefore the equation of the tangent is $y-\left(-\frac{1}{2}\right)=-(x-\left(-\frac{1}{2}\right))$ which simplifies to $y=-x-1$. A computer-generated plot visually confirms our computations.

Solution. 44.f

$$\begin{array}{rcll}
y^3+x^3+4xy &=& -4 & \Big| \text{ apply } \frac{\mathrm{d}}{\mathrm{d}x} \\
3y^2\dfrac{\mathrm{d}y}{\mathrm{d}x}+3x^2+4\left(y+x\dfrac{\mathrm{d}y}{\mathrm{d}x}\right) &=& 0 & \\
\dfrac{\mathrm{d}y}{\mathrm{d}x}\left(3y^2+4x\right) &=& -3x^2-4y & \\
\dfrac{\mathrm{d}y}{\mathrm{d}x} &=& \dfrac{-3x^2-4y}{3y^2+4x} & \Big| \text{ substitute } x=1, y=-1 \\
\dfrac{\mathrm{d}y}{\mathrm{d}x}_{|x=1,y=-1} &=& \dfrac{-3\left(1\right)^2-4\left(-1\right)}{3\left(-1\right)^2+4\left(1\right)} & \\
\dfrac{\mathrm{d}y}{\mathrm{d}x}_{|x=-\frac{1}{2},y=-\frac{1}{2}} &=& \dfrac{1}{7} &
\end{array}$$

Therefore the equation of the tangent is $y-(-1)=\dfrac{1}{7}(x-1)$ which simplifies to $y=\dfrac{x}{7}-\dfrac{8}{7}$. A computer-generated plot visually confirms our computations.

45. (a) If V is the volume of a cube with edge length x and the cube expands as time passes, find $\frac{dV}{dt}$ in terms of $\frac{dx}{dt}$.

answer: $3x^2\frac{dx}{dt}$

(b) Each side of a square is increasing at a rate of 1cm/s. At what rate is the area of the square increasing when the area of the square is 9 cm^2?

answer: $6\mathrm{cm}^2/\mathrm{s}$

(c) The radius of a ball is increasing at a constant rate of 5mm/s. At a point in time we measure the radius of the ball to be 5cm.

- How fast is the volume increasing at the time of our observation?
- How fast is the surface area increasing at the time of our observation?

(d) At a point in time, we measure the surface area of a ball to be increasing at a rate of $5\mathrm{cm}^2/\mathrm{s}$ and the radius of the ball to be 5cm.

- How fast is the volume increasing at the time of our observation?
- How fast is the radius increasing at the time of our observation?

(e) At a point in time, we measure the volume of a ball to be increasing at a rate of $5\mathrm{cm}^3/\mathrm{s}$ and the radius of the ball to be 5cm.

- How fast is the radius increasing at the time of our observation?
- How fast is the surface area increasing at the time of our observation?

(f) A 5m long ladder is leaning on a vertical wall. The base of the ladder is 4m away from the wall. At the moment of observation, the top end of the ladder is sliding downwards along the wall at a speed of 10 cm/s.

- At the time of observation, how fast is the bottom end of the ladder sliding away from the wall
- At the time of observation, how fast is the midpoint of the ladder moving away from the wall?
- At the time of observation, how fast is the midpoint of the ladder moving downwards towards the ground?

(g) A street light is mounted at the top of a 4m tall pole. A woman 160 cm tall walks away from the pole at a speed of 5km/h along a straight path. How fast is the tip of her shadow moving when she is 10m from the pole?

(h) A ship is pulled into a dock. On the dock, the rope is pulled by a pulley that is 1m lower than the mooring point on the ship. If the rope is pulled in at a constant rate of 10cm/s, how fast is the mooring point approaching the dock when it is 10m from the dock?

(i) A Ferris wheel with a radius of 10m is rotating at a rate of one revolution every 2 minutes. How fast is a riding rising when his seat is 16 m above ground level?

(j) The minute hand on a watch is 10mm long and the hour hand is 5mm long. How fast is the distance between the tips of the hands changing at two o'clock?

answer: (not proofread) $-\frac{55}{42}\sqrt{21}\pi$ mm/h

Solution. 45.a. The volume V is given by $V = x^3$, therefore

$$\frac{\mathrm{d}V}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(x^3\right) = 3x^2\frac{\mathrm{d}x}{\mathrm{d}t}.$$

Solution. 45.b The area A of the square is given by $A = x^2$, therefore

$$\frac{\mathrm{d}A}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(x^2\right) = 2x\frac{\mathrm{d}x}{\mathrm{d}t}.$$

When $A = 9\mathrm{cm}^2$, $x = 3\mathrm{cm}$ $(= \sqrt{9\mathrm{cm}^2})$, and so $\frac{\mathrm{d}A}{\mathrm{d}t}_{|x=3, \frac{\mathrm{d}x}{\mathrm{d}t}=1} = 2\cdot 3\mathrm{cm}\cdot 1\mathrm{cm/s} = 6\mathrm{cm}^2/\mathrm{s}$.

Solution. 45.d Let S denote the surface area and r the radius of the ball. Then $S = 4\pi r^2$. Let the point of time be t_0. We are given that $\frac{\mathrm{d}S}{\mathrm{d}t}_{|t=t_0} = 5\mathrm{cm}^2/\mathrm{s}$. On the other hand,

$$\begin{array}{rcl}\frac{\mathrm{d}S}{\mathrm{d}t} & = & \frac{\mathrm{d}}{\mathrm{d}t}\left(4\pi r^2\right) = 8\pi r\frac{\mathrm{d}r}{\mathrm{d}t}\\ \frac{1}{8\pi}\frac{\mathrm{d}S}{\mathrm{d}t} & & \end{array}$$

Solution. 45.j Let the angle between the two arrows be θ. The cosine law states that for a triangle with angle θ and sides a, b, c we have that $c^2 = a^2 + b^2 - 2ab\cos\theta$ (where c is the length of the side opposite to the angle θ).

Then by the cosine law, the distance between the tips of the two hands is 1

$$y = \sqrt{5^2 + 10^2 + 2\cdot 5\cdot 10\cos\theta} = \sqrt{125 + 100\cos\theta}.$$

The short hand makes 1 full revolution every 12 hours, and the long hand makes 1 full revolution every 1 hour. Therefore the angle θ measured from the small hand to the long hand changes at the (constant) rate of $\frac{11}{12}$ revolutions per hour, or what is the same, at the rate $\frac{\mathrm{d}\theta}{\mathrm{d}t} = \frac{11}{12}(2\pi) = \frac{11}{6}\pi$.

The problem asks us to compute $\frac{dy}{dt}$ at two o'clock, i.e., at $t = 2$. This is a straightforward computation using the chain rule:

$$\begin{array}{rcll}\frac{\mathrm{d}y}{\mathrm{d}t} & = & \frac{1}{2}\frac{-100\sin\theta}{\sqrt{125+100\cos\theta}}\frac{\mathrm{d}\theta}{\mathrm{d}t} & \Big| \text{ at 2 o'clock } \theta = \frac{\pi}{3} \text{ and } \frac{\mathrm{d}\theta}{\mathrm{d}t} = \frac{11}{6}\pi\\ \frac{\mathrm{d}y}{\mathrm{d}t}_{|t=2} & = & \frac{1}{2}\frac{-100\sin\frac{\pi}{3}}{\sqrt{125+100\cos\frac{\pi}{3}}}\frac{11\pi}{6} = -\frac{55}{42}\sqrt{21}\pi & .\end{array}$$

The measurement unit of speed is mm/hour, so the distance is changing at the rate of $-\frac{55}{42}\sqrt{21}$ mm/hour.

46. Use the Intermediate Value theorem and the Mean Value Theorem/Rolle's Theorem to prove that the function has **exactly one** real root.

(a) $f(x) = x^3 + 4x + 7$.

(b) $f(x) = x^3 + x^2 + x + 1$.

(c) $f(x) = \cos^3\left(\frac{x}{3}\right) + \sin x - 3x$.

Solution. 46.a. $f(-2) = -9$ and $f(1) = 12$. Since $f(x)$ is continuous and has both negative and positive outputs, it must have a zero. In other words, for some c between -2 and 1, $f(c) = 0$. If there were solutions $x = a$ and $x = b$, then we would have $f(a) = f(b)$, and Rolle's Theorem would guarantee that for some x-value, $f'(x) = 0$. However, $f'(x) = 3x^2 + 4$, which always positive and therefore is never 0. Therefore there cannot be 2 or more solutions.

The above can be stated informally as follows. Note that $f'(x) = 3x^2 + 4$, which is always positive. Therefore, the graph of f is increasing from left to right. So once the graph crosses the x-axis, it can never turn around and cross again, so there can only be a single zero (that is, a single solution to $f(x) = 0$).

Solution. 46.c. $f(5) = \cos^3\left(\frac{5}{3}\right) + \sin 5 - 15 \le 2 - 15 = -13 < 0$ (because $\cos a, \sin b \in [-1, 1]$ for arbitrary a, b). Similarly $f(-5) = \cos^3\left(-\frac{5}{3}\right) + \sin(-5) + 15 \ge 15 - 2 > 0$. Therefore by the Intermediate Value Theorem $f(x) = 0$ has at least one solution in the interval $[-5, 5]$.

Suppose on the contrary to what we are trying to prove, $f(x) = 0$ has two or more solutions; call the first 2 solutions a, b. That means that $f(a) = f(b) = 0$, so by the Mean value theorem, there exists a $c \in (a, b)$ such that $f'(c) = (f(a) - f(b))/(a - b) = (0 - 0)/(a - b) = 0$. On the other hand we may compute:

$$f'(x) = -3 + \cos x - \cos^2\left(\frac{x}{3}\right)\sin\left(\frac{x}{3}\right) \le -1 < 0,$$

where the first inequality follows from the fact that $\sin x, \cos x \in [-1, 1]$. So we got that $f'(c) = 0$ for some c but at the same time $f'(x) < 0$ for all x, which is a contradiction. Therefore $f(x) = 0$ has exactly one solution.

47. Use the Intermediate Value theorem and the Mean Value Theorem/Rolle's Theorem to prove that the function has **exactly one** real root.

    (a) $x^5 + 7x = 2$.

    (b) $x^7 + x^5 + x^3 = 3$.

    (c) $2x - 1 = \sin x$.

    (d) $e^x + 2x = 3$.